

Simulating First Objects

Vasilii Pustovoit, 2023/11/13



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TORONTO



CITA
ICAT

Canadian Institute for
Theoretical Astrophysics
L'institut Canadien
d'astrophysique théorique

Star populations

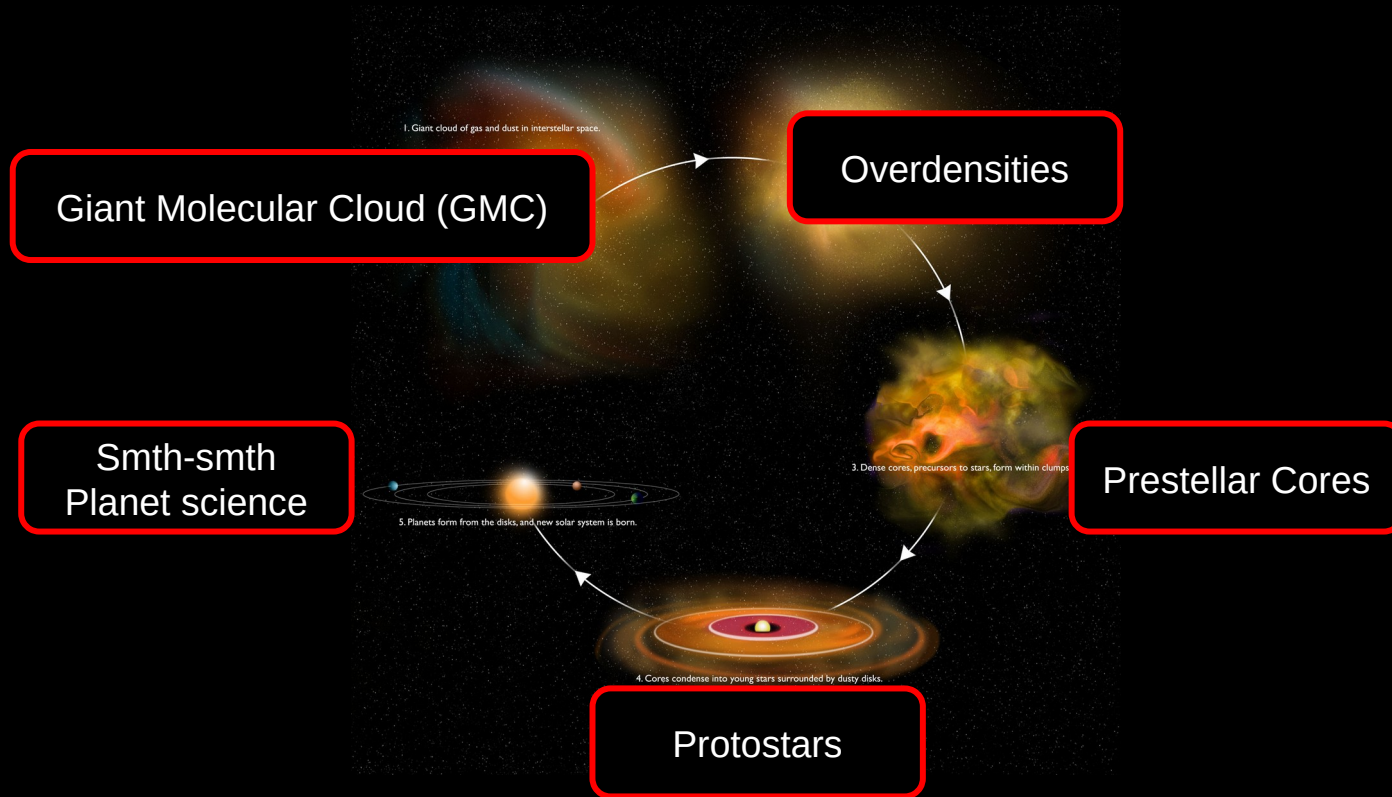
- Population I stars (PopI)
 - Subsequent generations of stars
 - High metallicity (0.1 to 3 solar)
 - Wide range of masses
 - Observed (duh)
- Population II stars (PopII)
 - Second generation of stars
 - Low metallicity (10^{-5} to 10^{-1} solar)
 - More massive than PopI
 - Observed
- Population III stars (PopIII)
 - First generation of stars
 - Virtually 0 metallicity ($<10^{-5}$ solar)
 - Very massive (up to $10^3 M_{\text{sun}}$!)
 - Not observed (yet)

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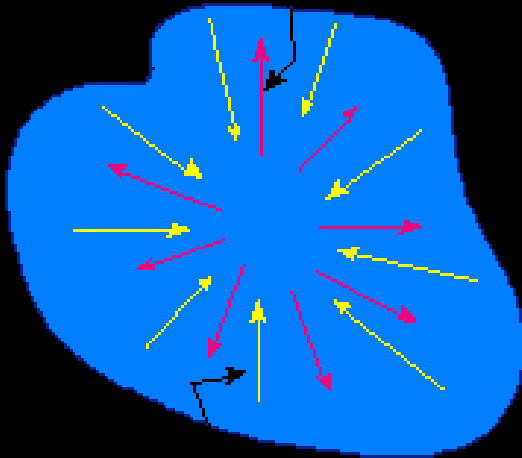
(First Stars)

PopI Star Formation 101



Popl Star Formation 101

Gas pressure attempting to
expand the cloud

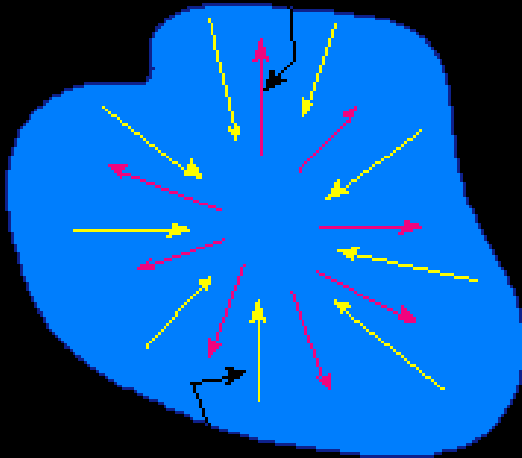


Gravitational force attempting
to collapse the cloud

Pressure vs Gravity

Popl Star Formation 101

Gas pressure attempting to
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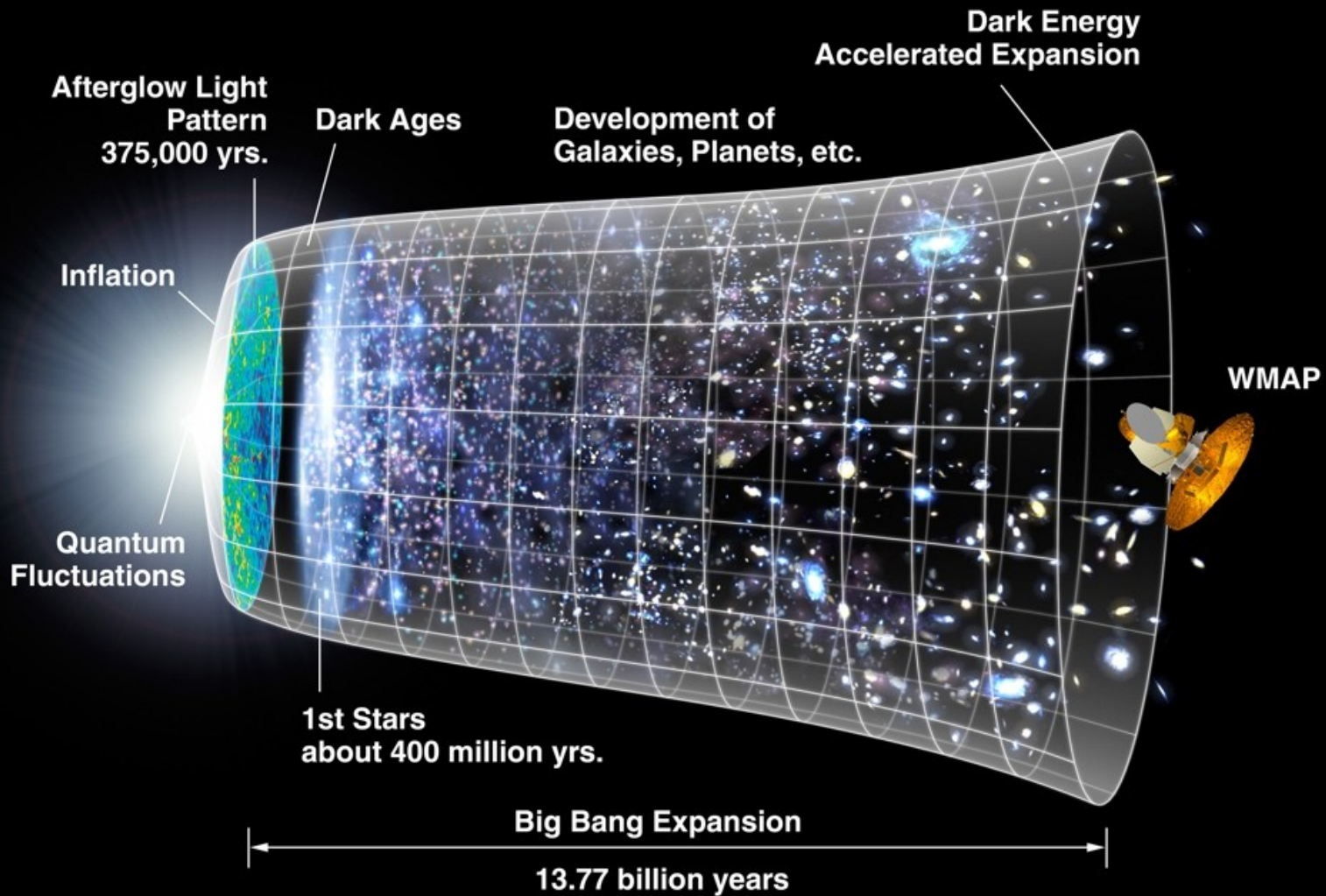
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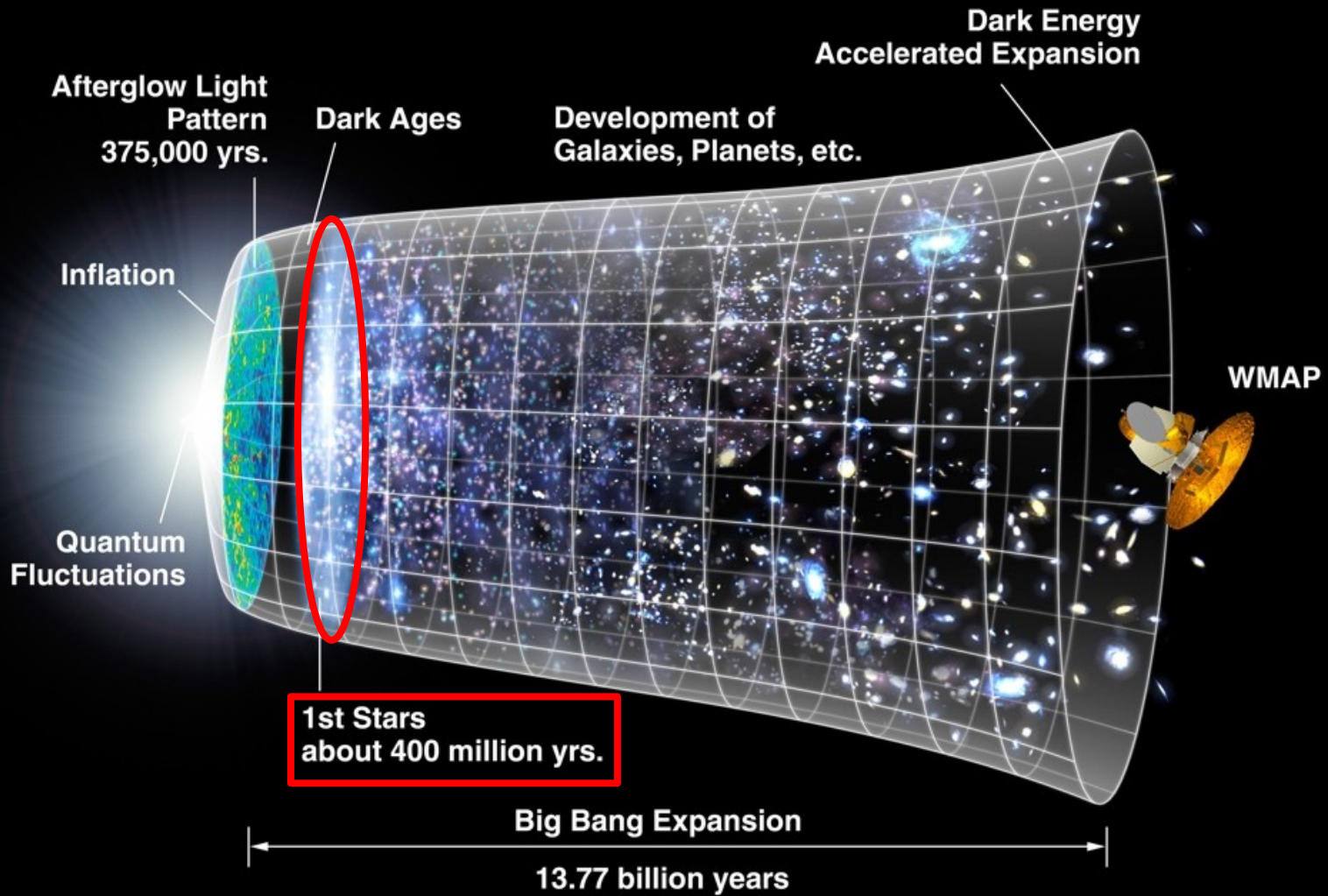
Pressure vs Gravity

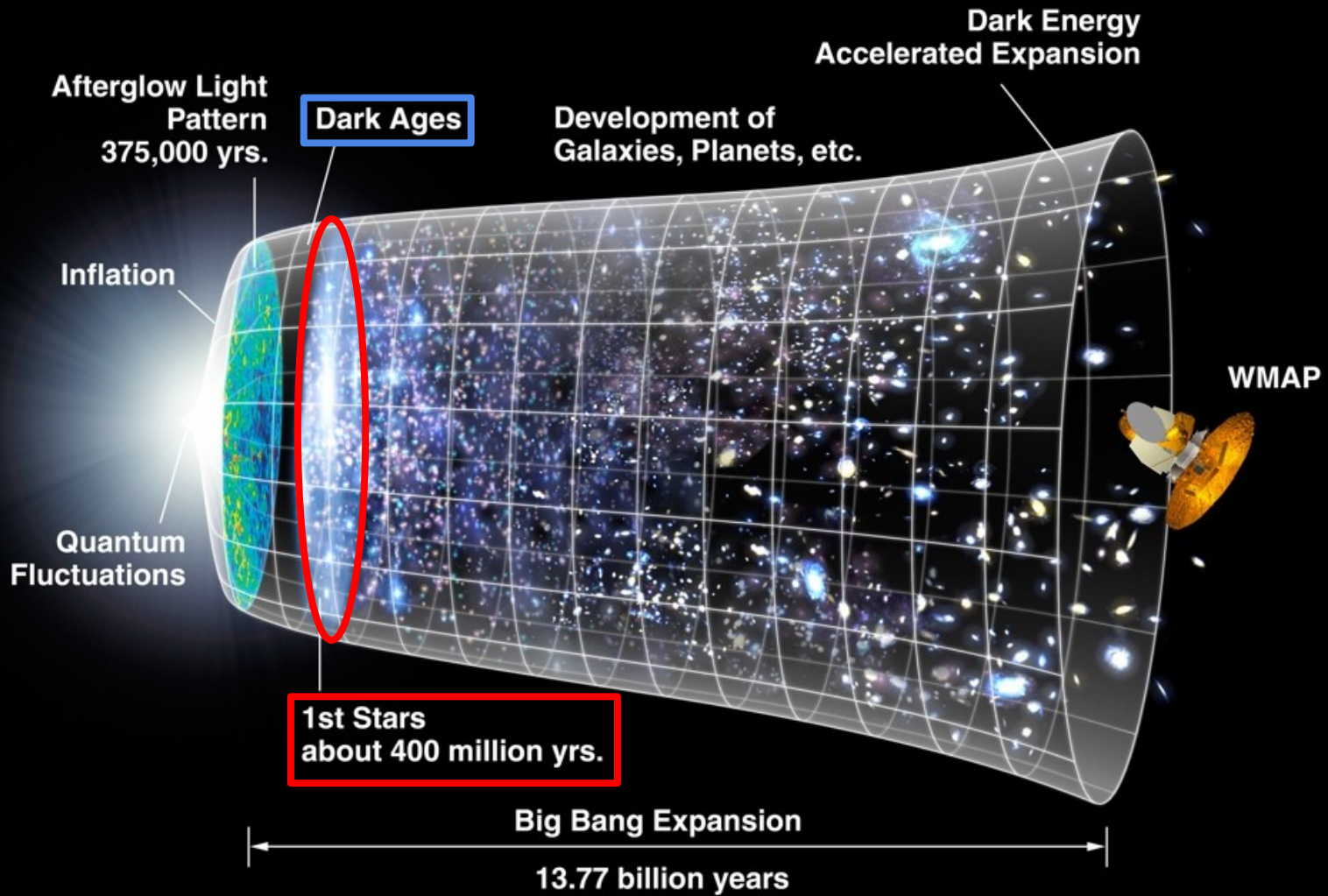
For gravitational collapse to happen:

$$t_{ff} < t_{sound} \quad \Leftrightarrow \quad F_G > F_{pres}$$

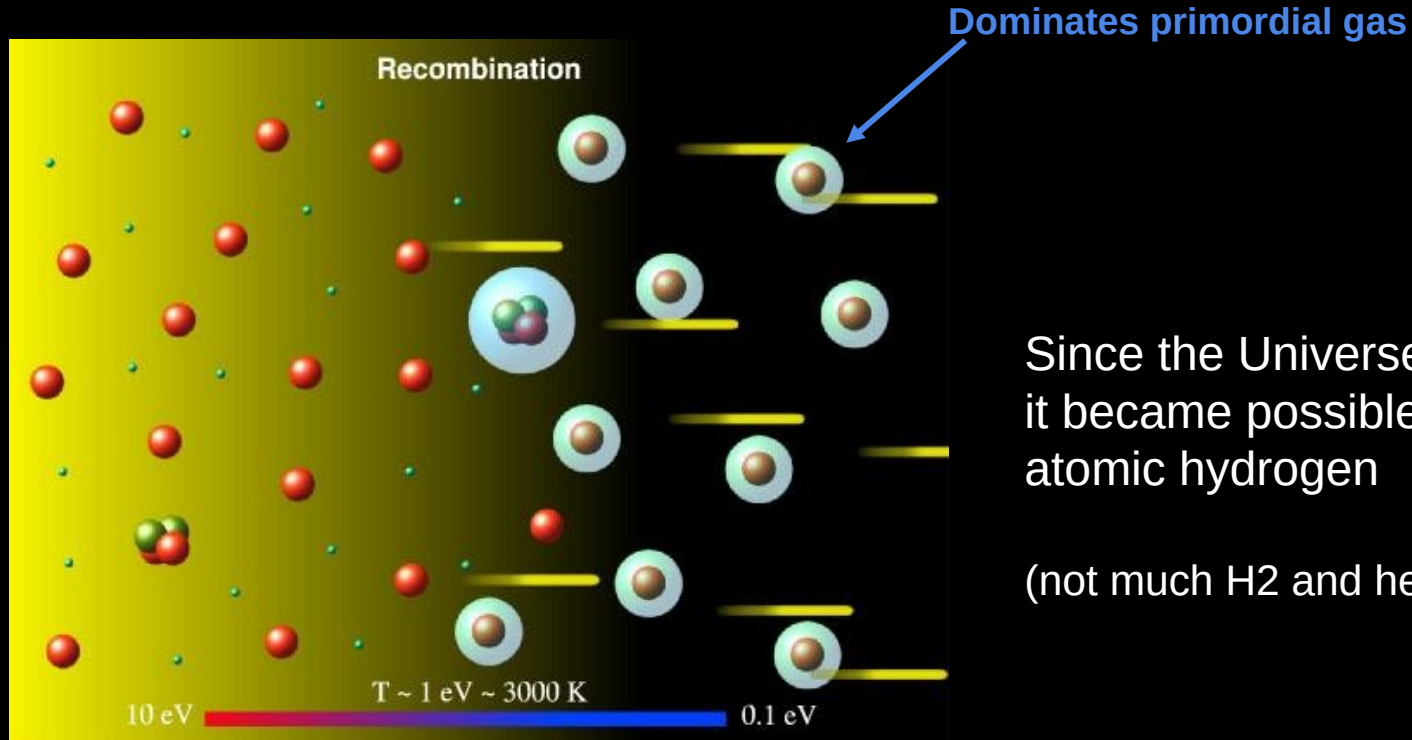
How is formation of PopIII stars different from that of PopI?







Cosmic Dark Ages and H



Since the Universe cooled down,
it became possible to produce
atomic hydrogen

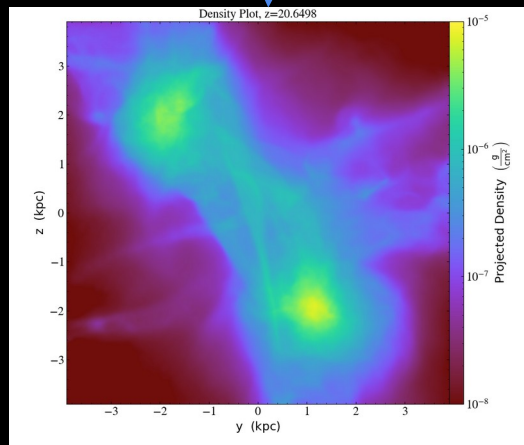
(not much H₂ and heavier elements)

Primordial vs Modern Star-Forming Regions

Less efficient at cooling!



H



Primordial “GMC”

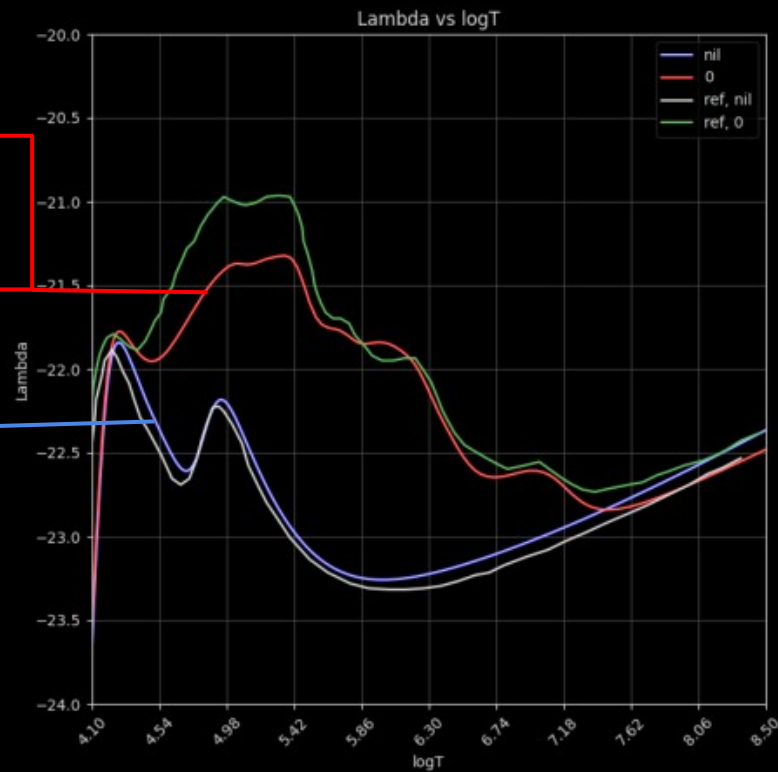
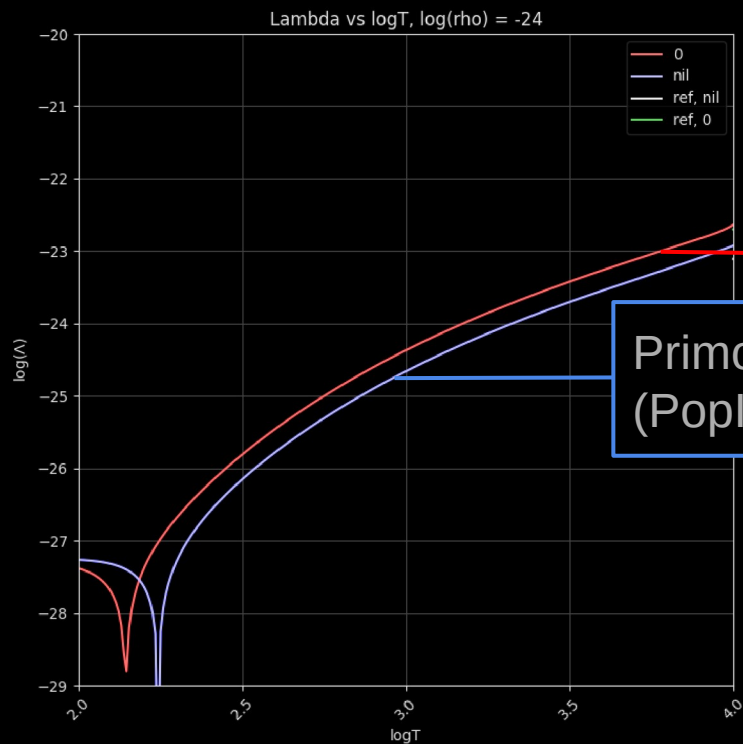


H₂



Present-day GMC

Cooling Functions



Reference (white, green): Sutherland & Dopita, 1993

How is formation of PopIII stars different from that of PopI?

How is formation of PopIII stars different from that of PopI?

- Because of **H** (high-z) **vs H₂** (low-z), PopIII stars are **much harder to cool** as they are being formed!
- **Metallicity** is also extremely important, and for high-z, it is ~ 0 . Thus no cooling due to metals and dust

Let's form some PopIII stars!

Questions to answer:

- What physics is relevant to consider when we talk about star formation at high redshift?
- What are the IMFs of PopII and PopIII stars?
- What are interesting effects in the binary population of PopIII stars?
- Can IMBHs (Intermediate Mass Black Holes) form in the Early Universe?
- What are paths for evolution of PopIII star remnants?
- What roles do these primordial objects play in reionization?

Along with other questions that arise on the way

Initial Conditions: What Do We Know?

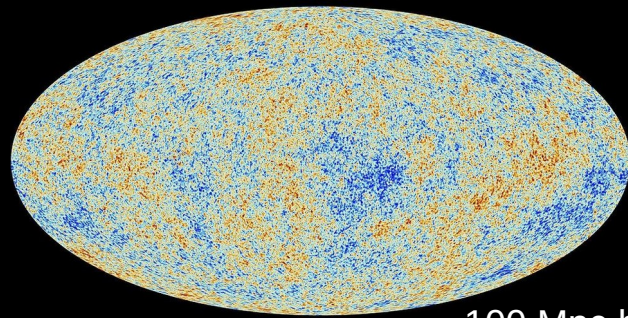
Not much...

Initial Conditions: What Do We Know?

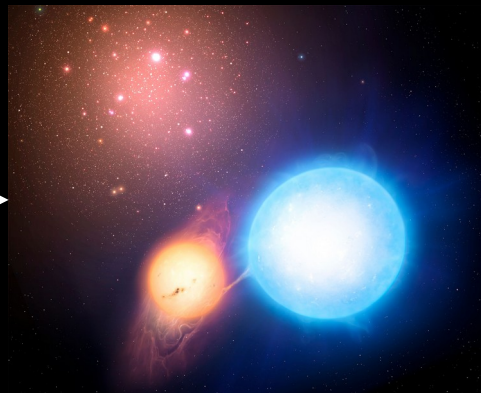
- We cannot observe PopIII stars (too small)
- We cannot observe clouds in which they form (still too small)
- We have no idea what IMF (Initial Mass Function) of PopIII **actually** is...
- But there is something we know **very well** in the Early Universe

Initial Conditions: What Do We Know?

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~100 Mpc box



<AU-sized objects

Big Issue: Multi-Scale Approach

How do we resolve small scales (stars) if we can only have large scales for initial conditions (CMB)?

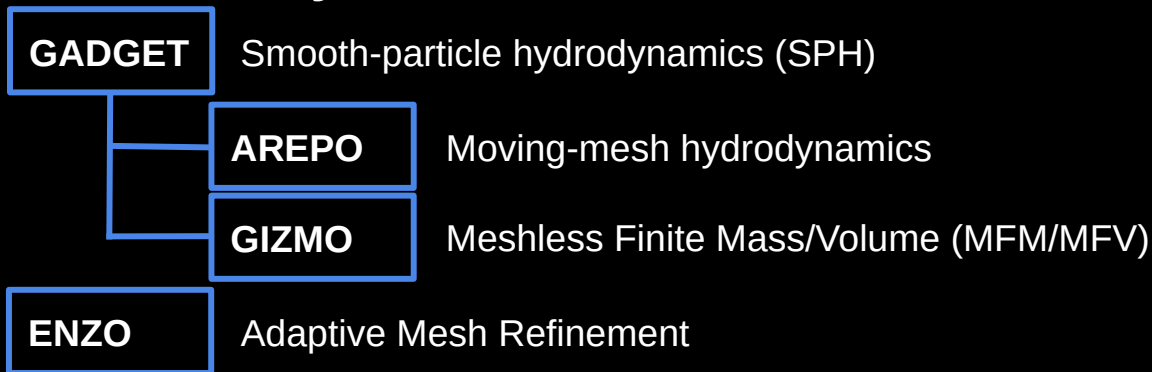
Possible Solution: Hydrosims

- Can be coupled to other physics
- Realistic fluid behavior
- High spatial resolution
- Pretty general
- Can do “zoom-in” simulations!

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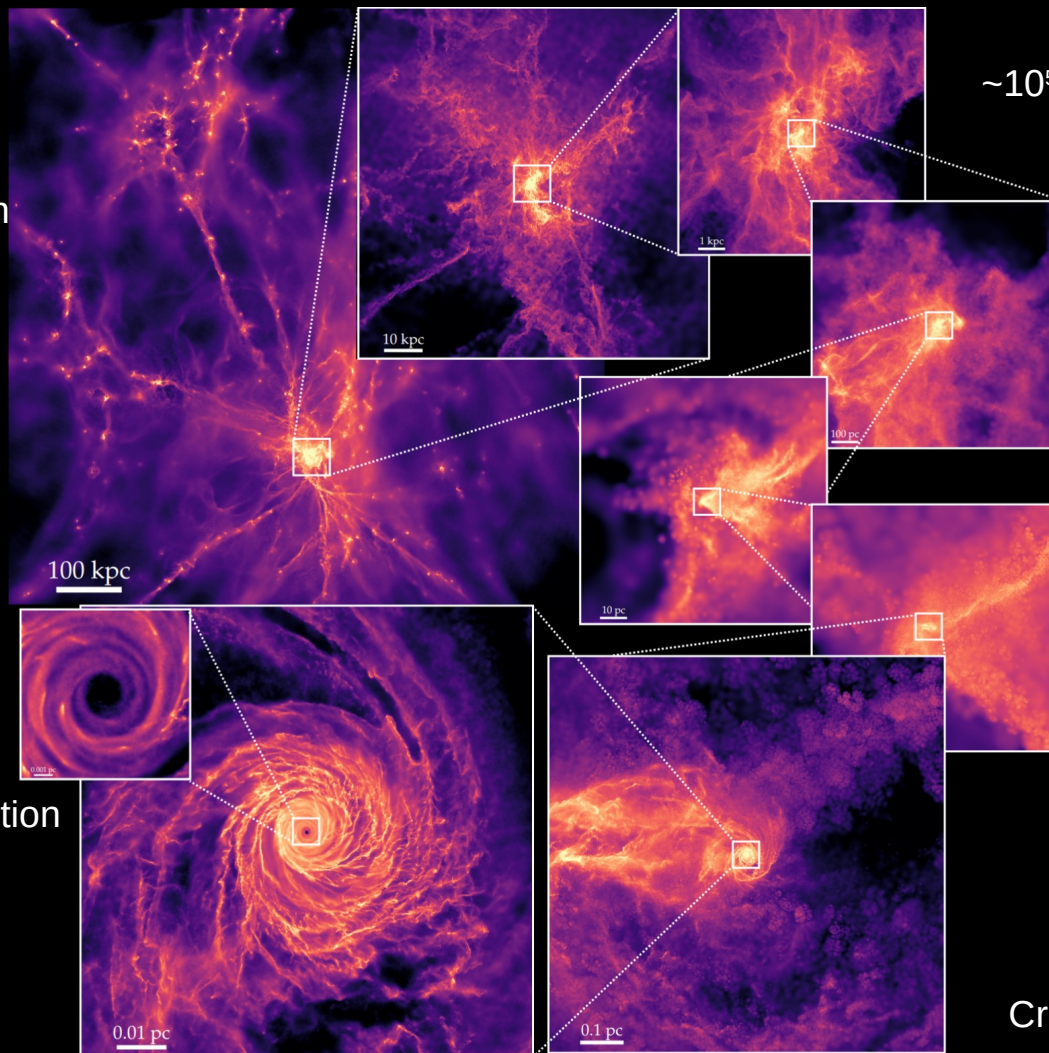
- Can be coupled to other physics
- Realistic fluid behavior
- High spatial resolution
- Pretty general
- Can do “zoom-in” simulations!

Which Hydrosim?



Has anyone done something similar?

$\sim 10^7$ Msun resolution



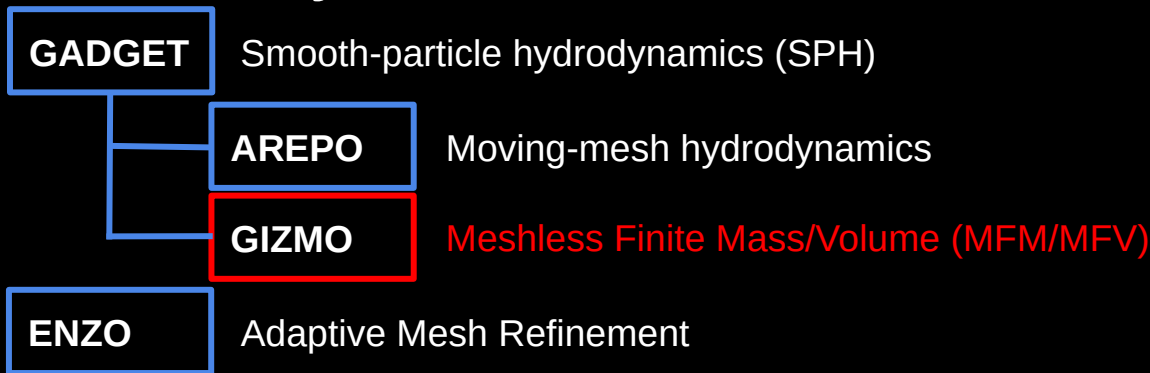
$\sim 10^{-3}$ Msun resolution

Credit: 2309.13115

Possible Solution: Hydrosims

- Can be coupled to other physics
- Realistic fluid behavior
- High spatial resolution
- Pretty general
- Can do “zoom-in” simulations!

Which Hydrosim?



Specifications of GIZMO: STARFORGE Module

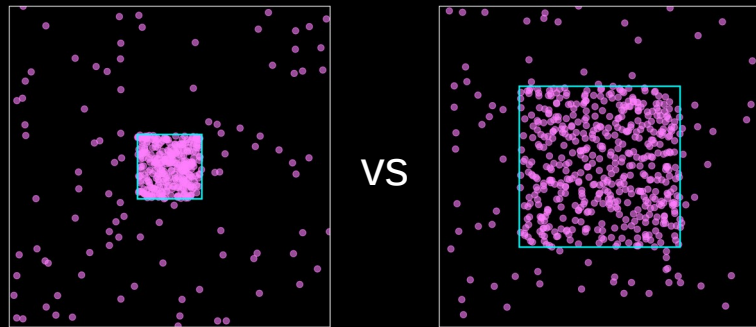
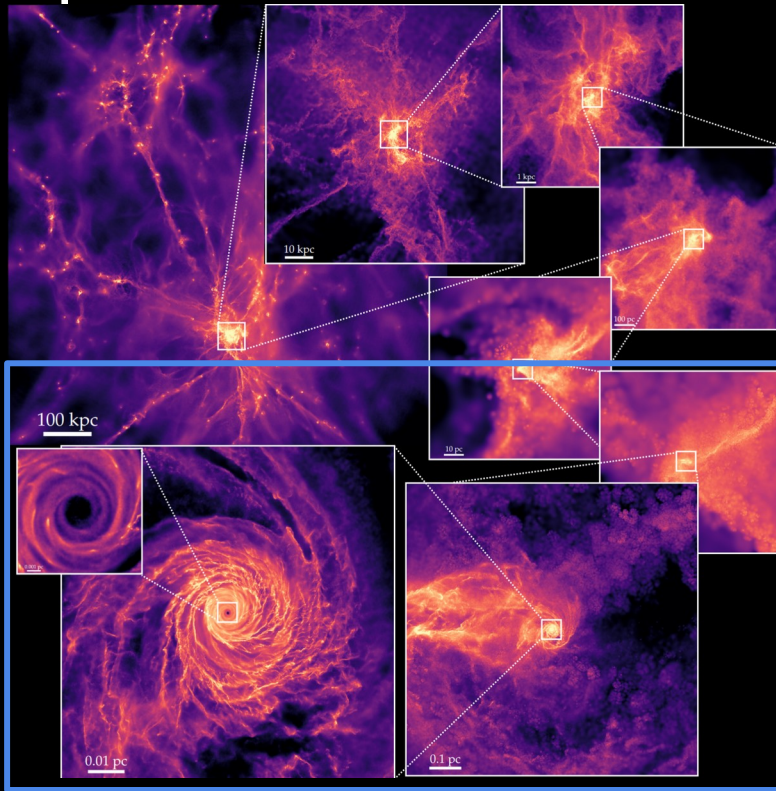
Has been used to resolve star formation in GMCs/near AGNs

Stars are accreting particles (No IMF is needed!)

Considers MHD, stellar feedback, etc.

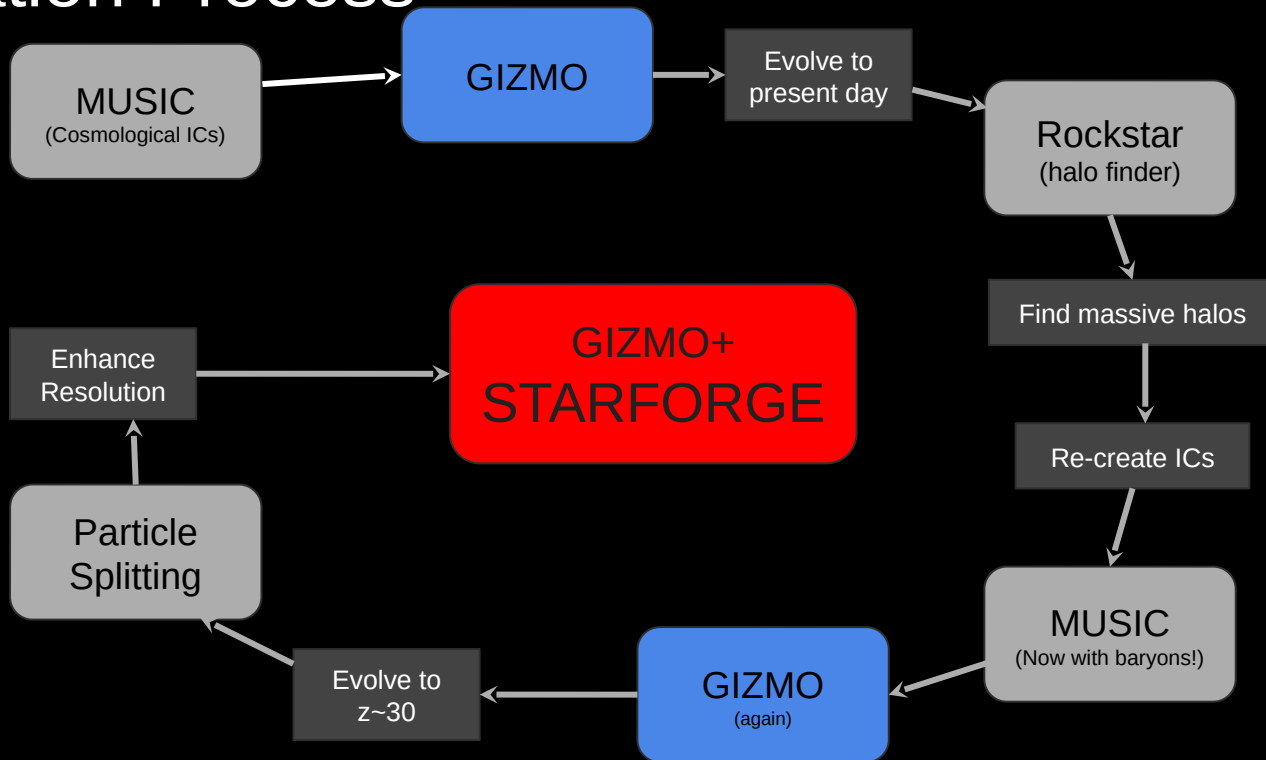


Specifications of GIZMO: Particle Splitting

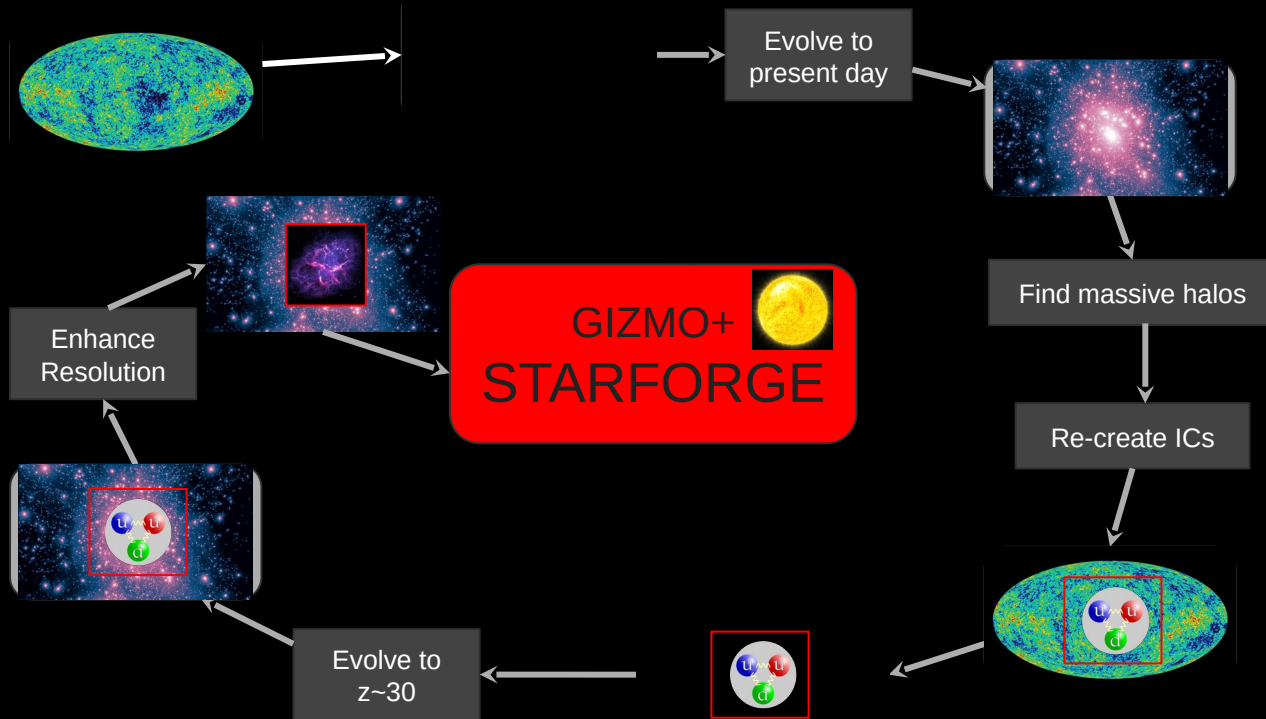


Zoom-in ICs do not produce enough resolution to resolve these scales

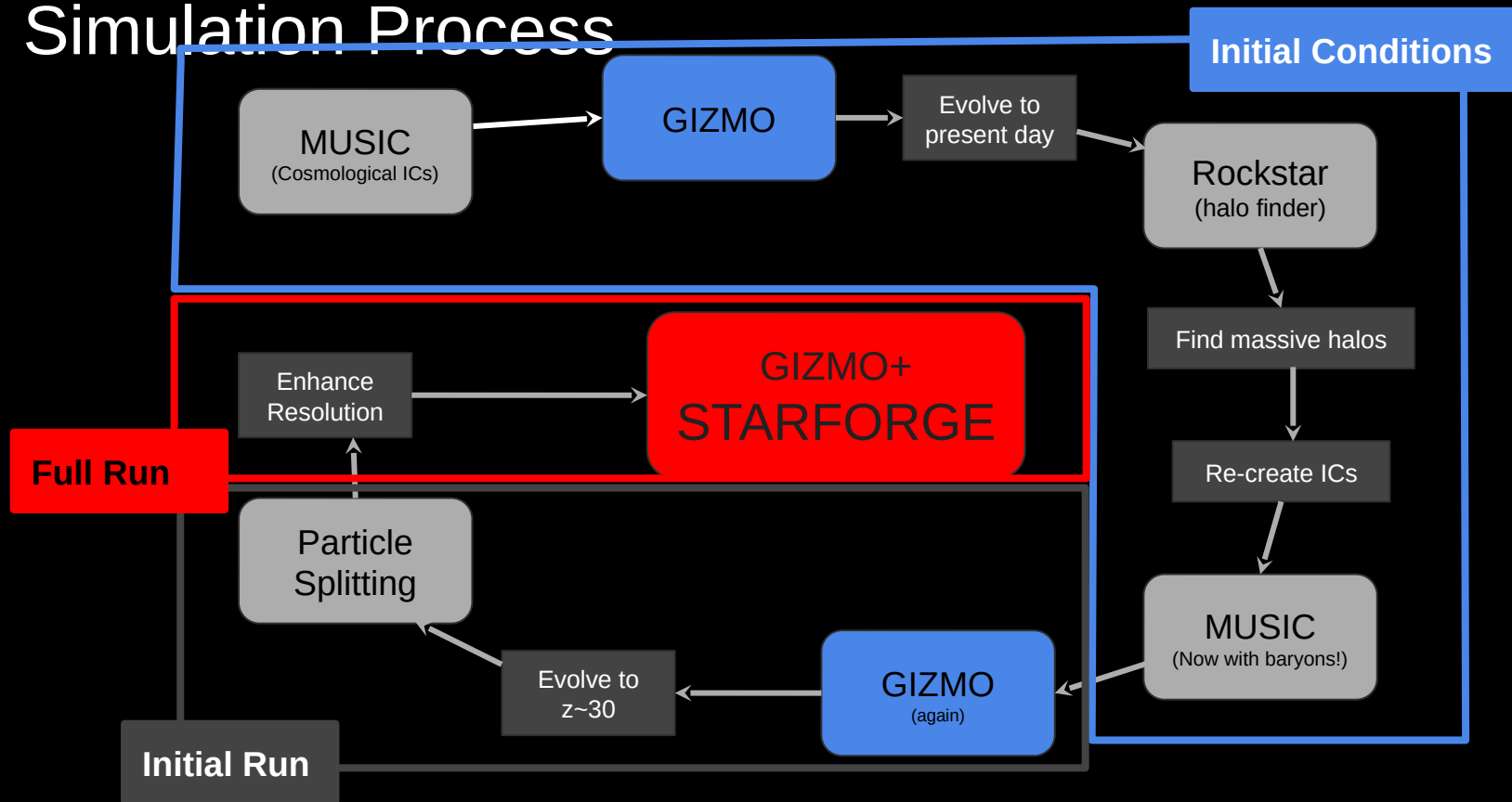
Simulation Process



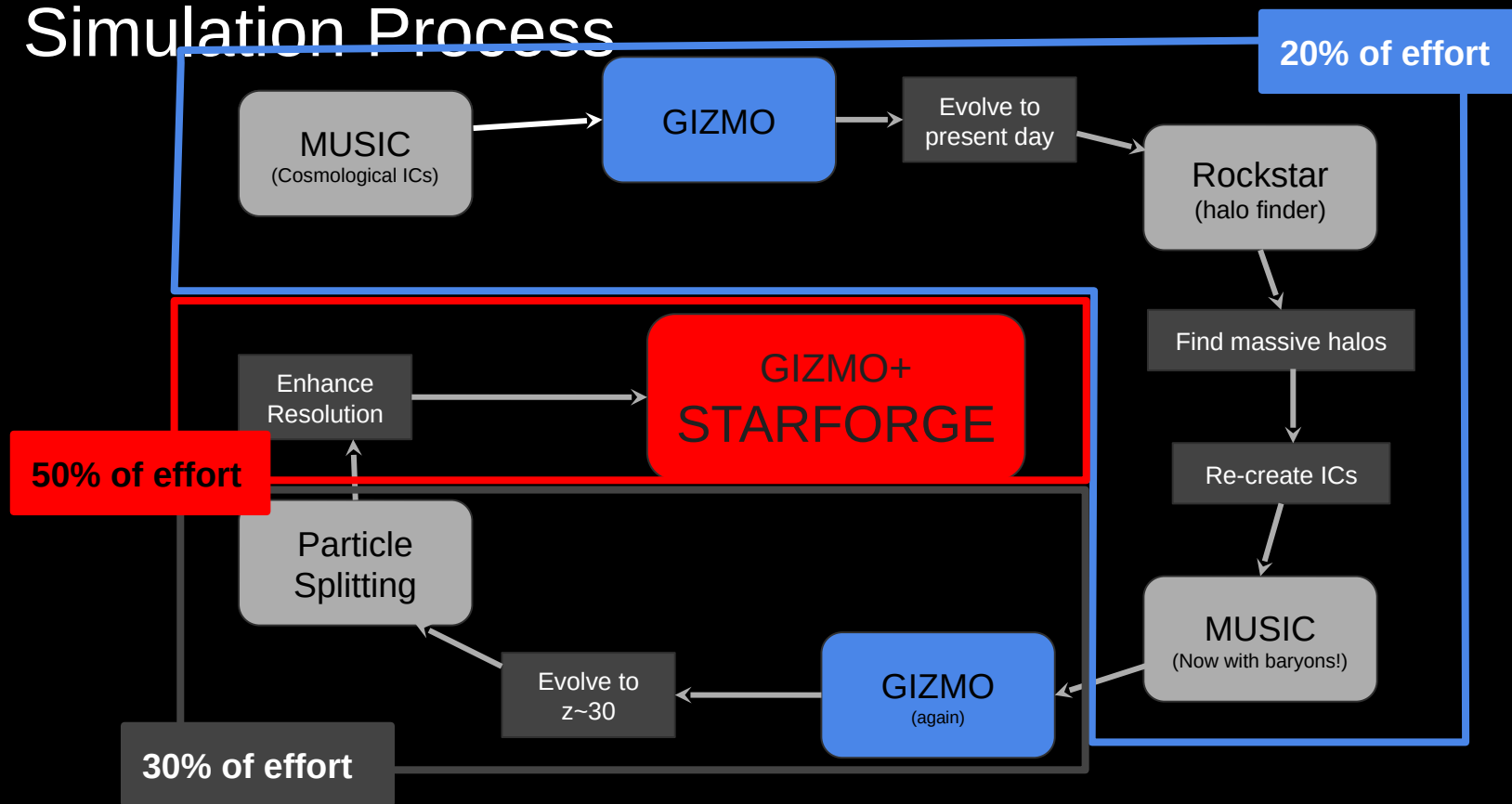
Simulation Process



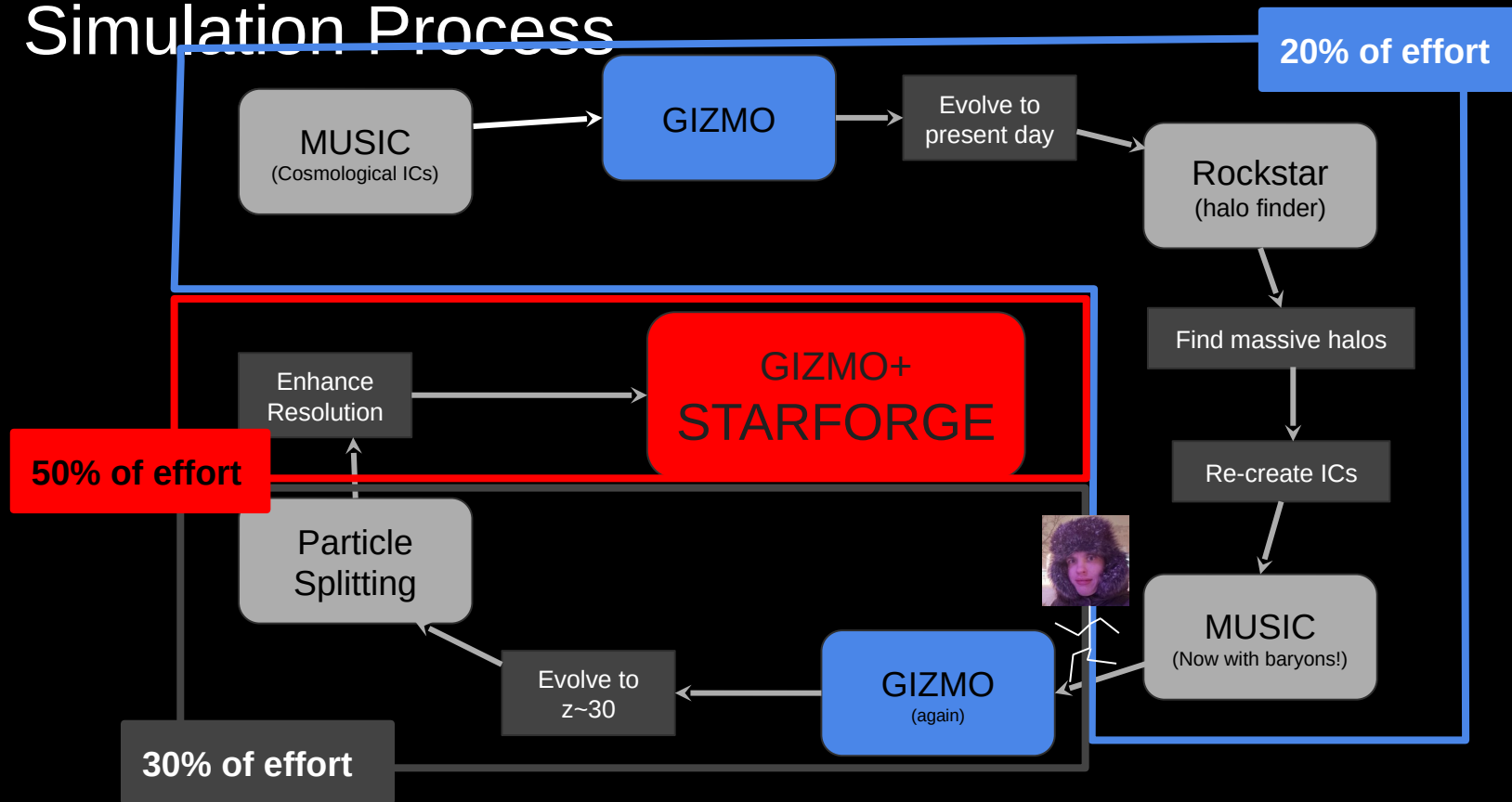
Simulation Process



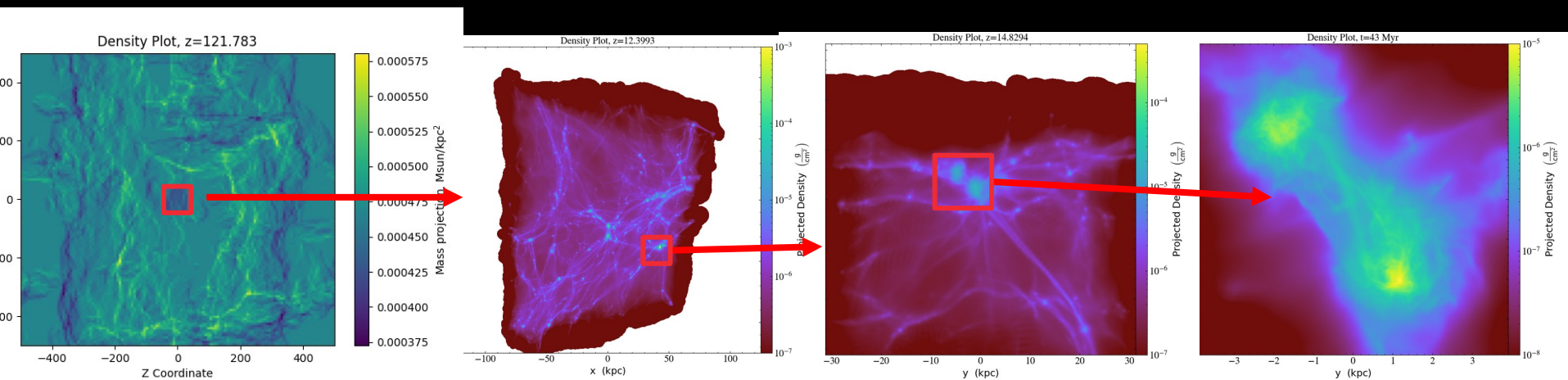
Simulation Process



Simulation Process



What we have so far...



Still not enough resolution for individual stars...

Current issues

- Picking the right bounding box for 2nd set of MUSIC ICs (90% solved)
- Cooling at high- z (Check that physics makes sense)
- Particle splitting (Main issue found, haven't been run yet)
- Physicality of particle splitting (More tests need to be ran)
- Do current ICs produce physical results? (In progress)

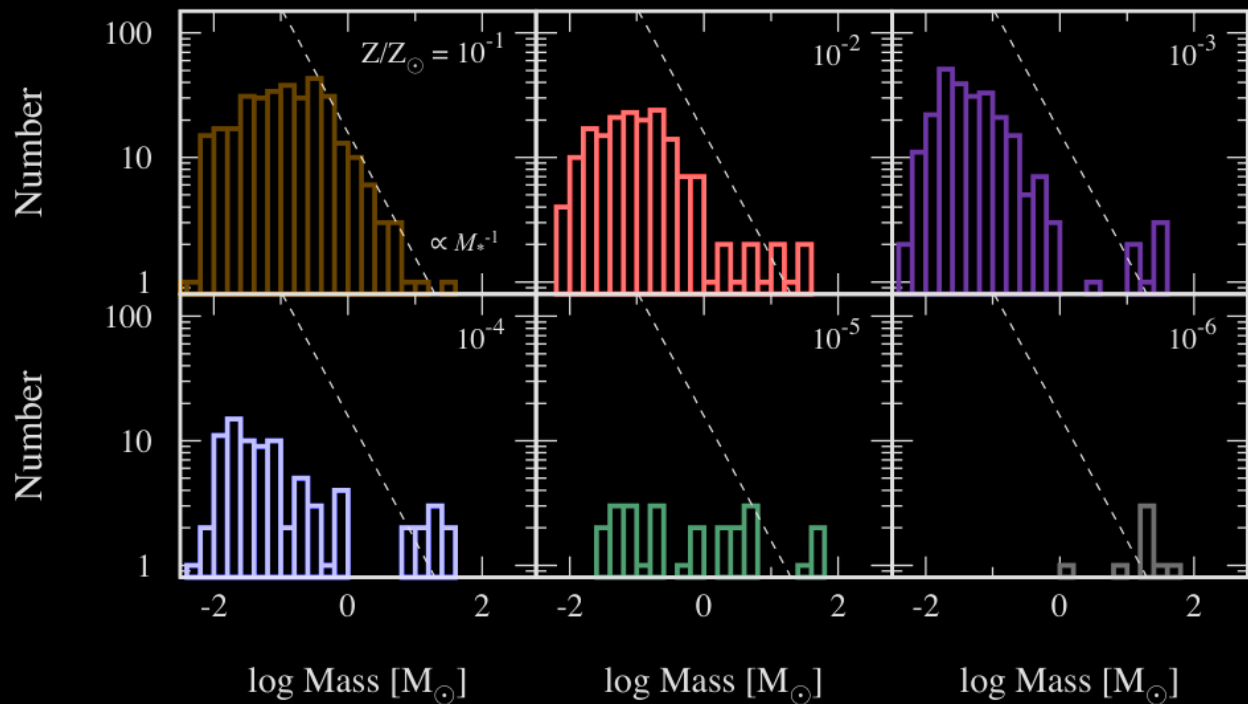
Timeline

Tasks	Start Date	End Date	2023				2024				2025				2026			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
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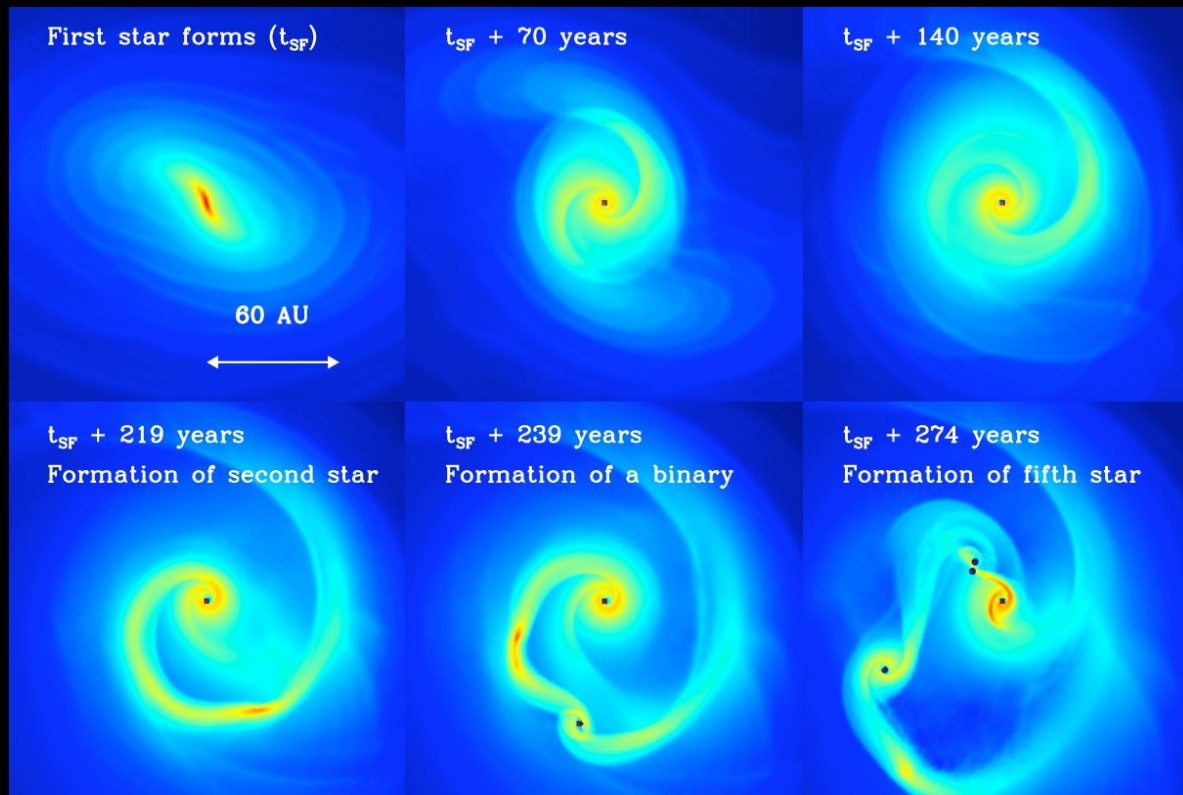
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Potential First Paper Material: IMF of First Stars



It is theorized that Pop III stars' IMF will be **logarithmically flat** (more high-mass stars compared to Pop I/II)

Potential First Paper Material: First Star Clusters



High- z accretion disks are highly susceptible to fragmentation

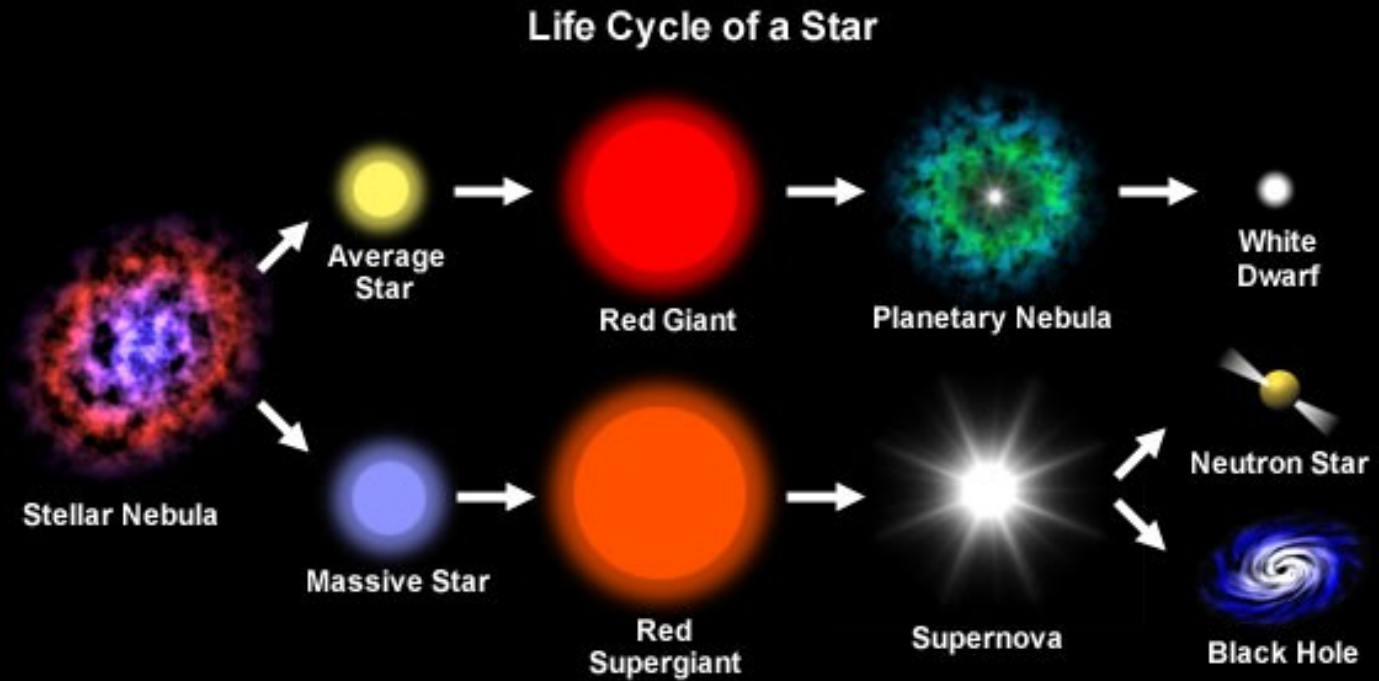
>>>

Pop III are expected to form in clusters!

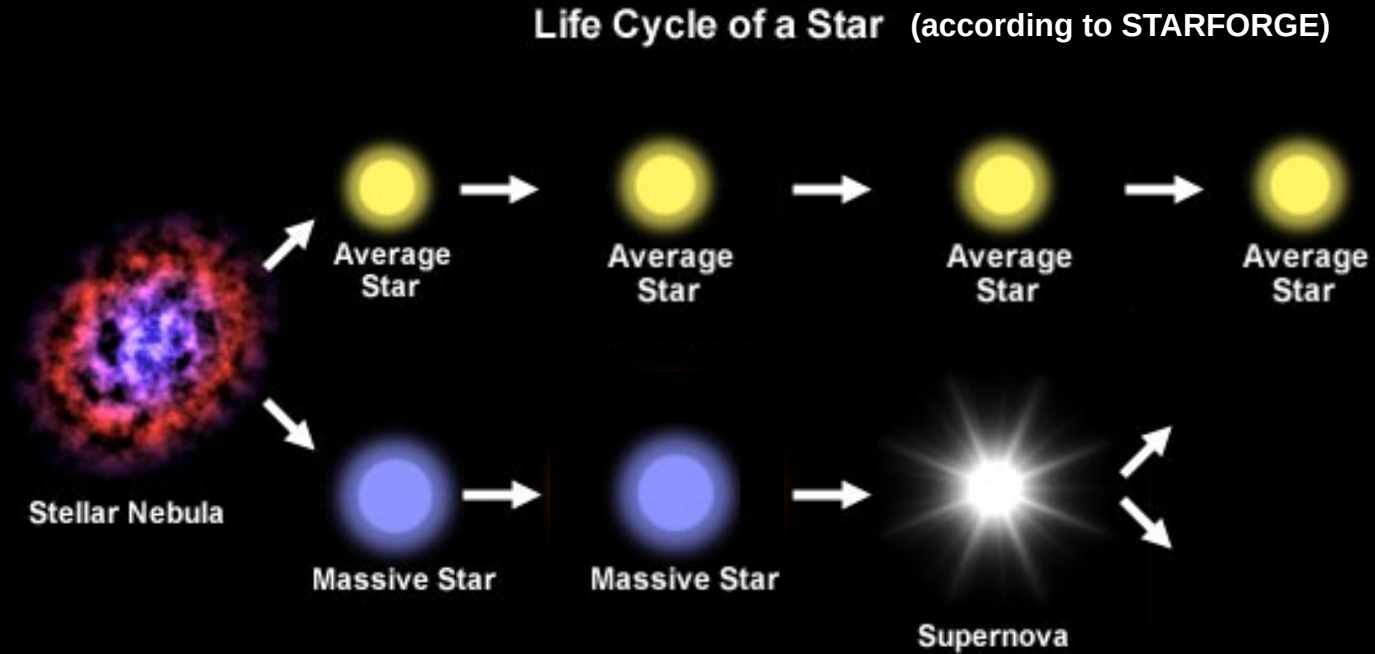
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Issues with STARFORGE



Issues with STARFORGE



Issues with STARFORGE

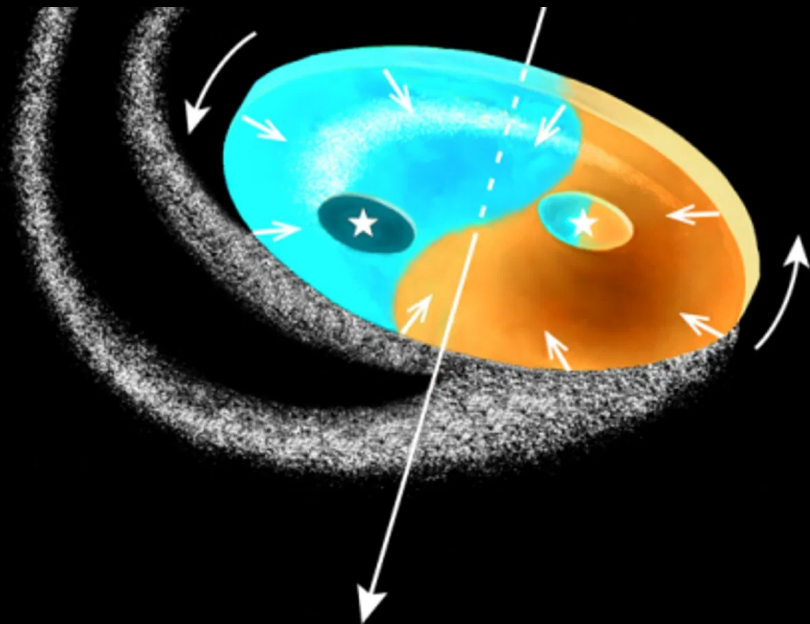
Life Cycle of a Star (according to STARFORGE)



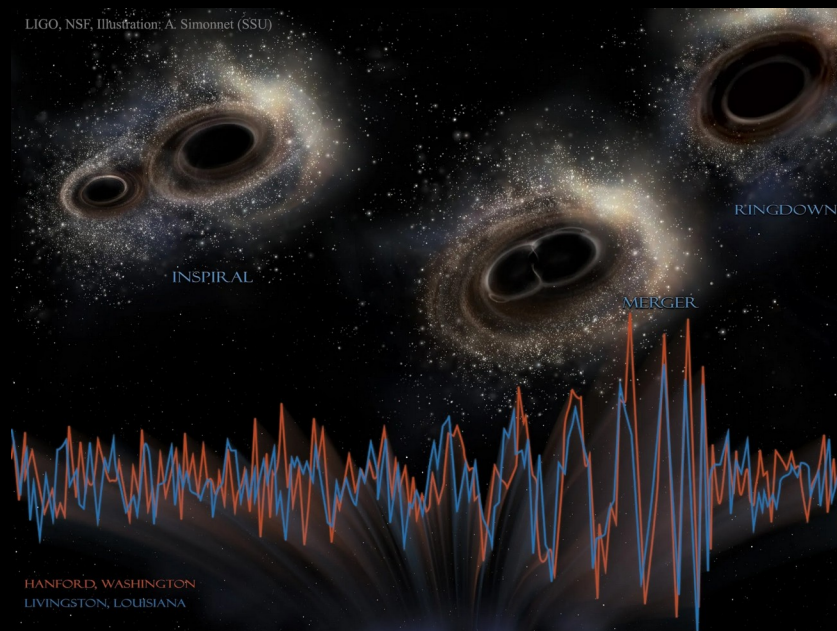
**Will have to code
this myself!**

Potential Second Paper Material: Binaries and remnants

It is theorized that Pop III stars might evolve into High Mass X-ray Binaries (HMXBs)



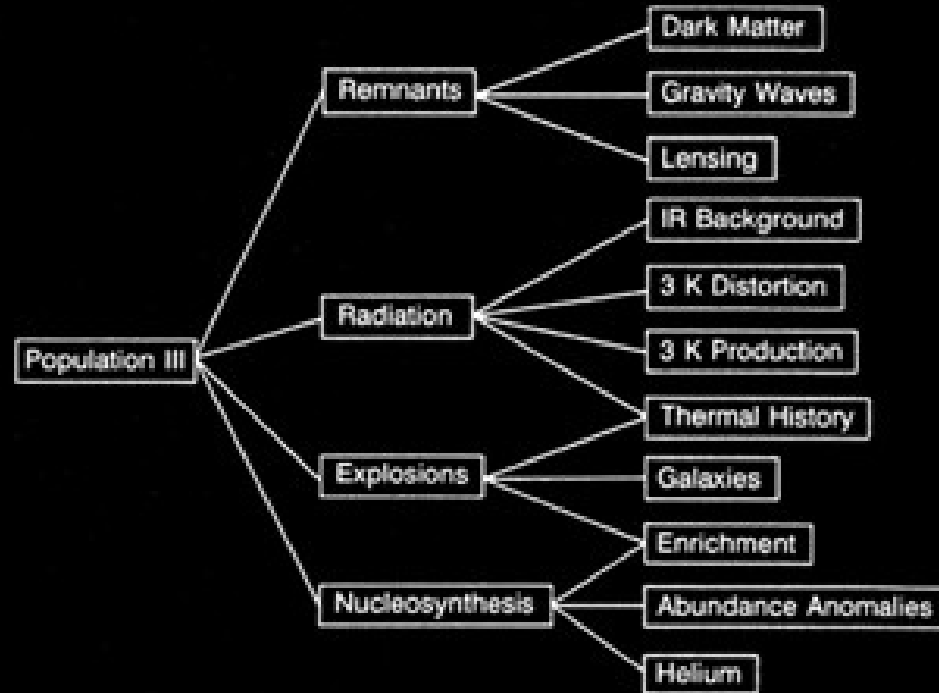
Tight binaries might become detectable BBH mergers



Timeline

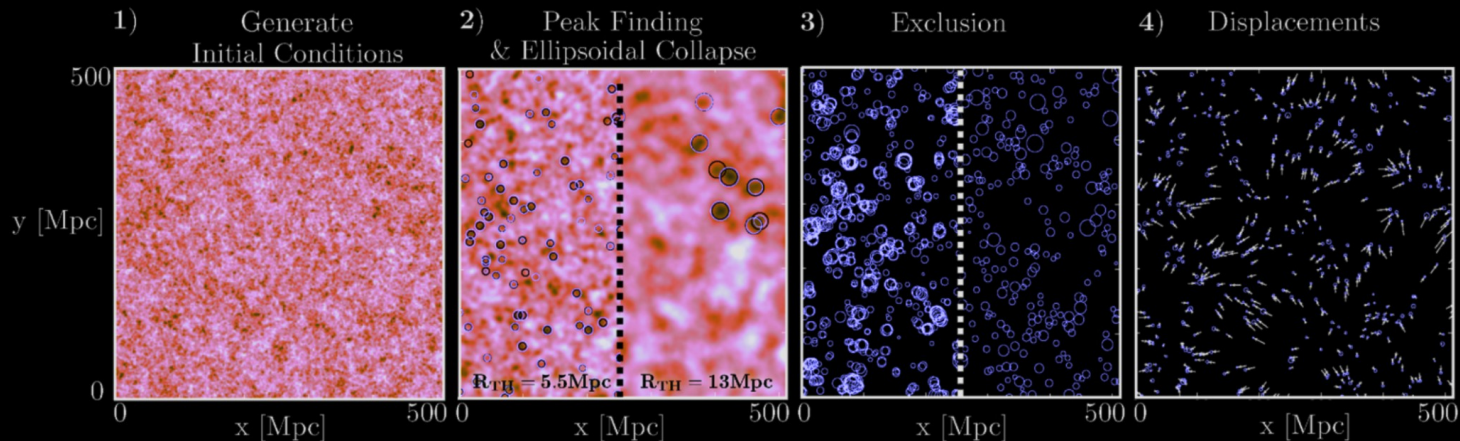
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Potential Third Paper Material: Cosmological Impact



Potential Third Paper Material: Connection to Peak Patch!

- Peak Patch generates mock catalogues of DM halos
- In order to go from DM to baryons, need to know transfer function at high z - depends on star formation



Summary

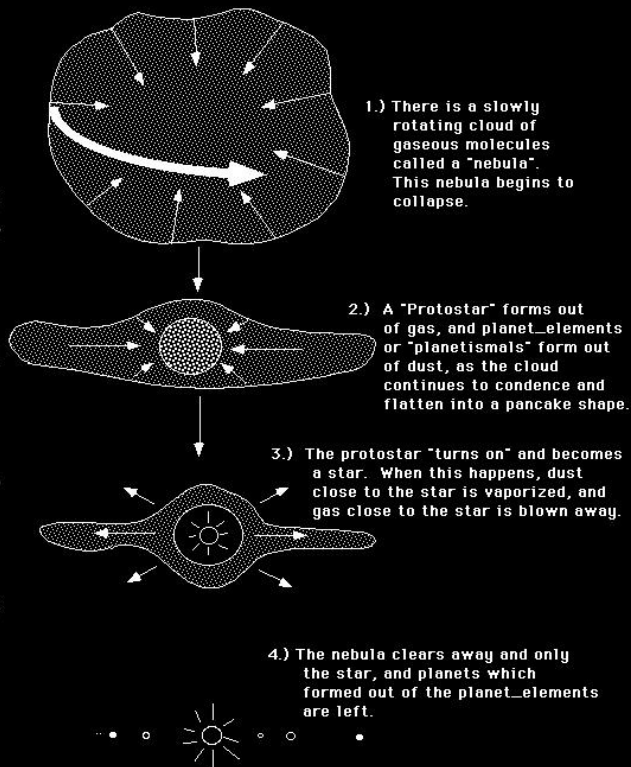
- Star formation at high- z is very different from low- z
- Need to run zoom-in cosmological hydrosim to get physical results
- For that, it's very important to understand the physics of the processes to pick appropriate modules
- LOTS of physics to explore!

Thank you!

Appendix A: Details on Star Formation

Cloud Collapse and Star/Planet Formation

- o Jeans cloud collapse equations describe the conditions required for an ISM cloud to collapse.
- o As a cloud collapses, central temperature increases.
- o This is accompanied by spinning-up of the central star (to conserve AM).
- o Disk also flattens into an oblate spheroid.



Appendix B: MUSIC Comparison

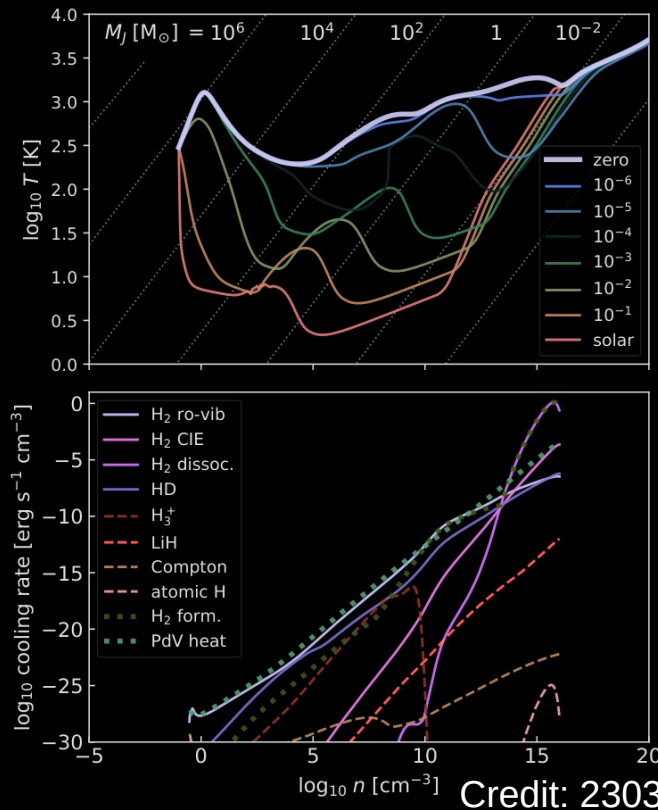
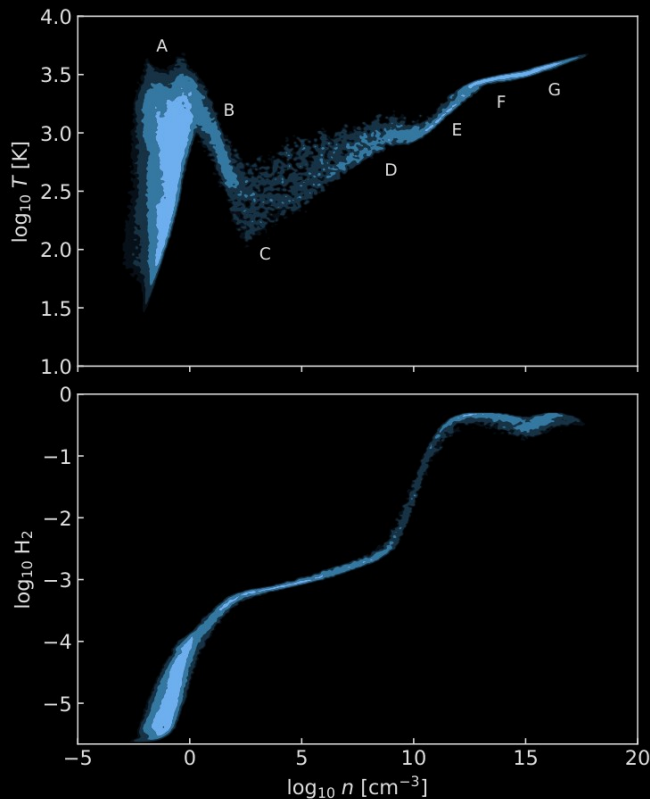
Feature / Code	MUSIC	N-GenIC	2LPT	GRAFIC	IC_gen
Multi-Scale ICs	Yes	No	No	No	Yes
Zoom-In Capabilities	Yes	No	No	No	Limited
Parallel Computation	Yes	Yes	Yes	No	Yes
2LPT Support	Yes	No	Yes	No	No
Custom Power Spectrum	Yes	No	No	Yes	Yes
Baryon Physics	Yes	No	No	No	No
Compatibility	High	Medium	Medium	Low	Medium
User-Friendly Interface	Medium	Low	Low	Low	Medium
Comments	Industry Standard				Comes with SWIFT

Table 1: Comparison of features across various initial condition generators

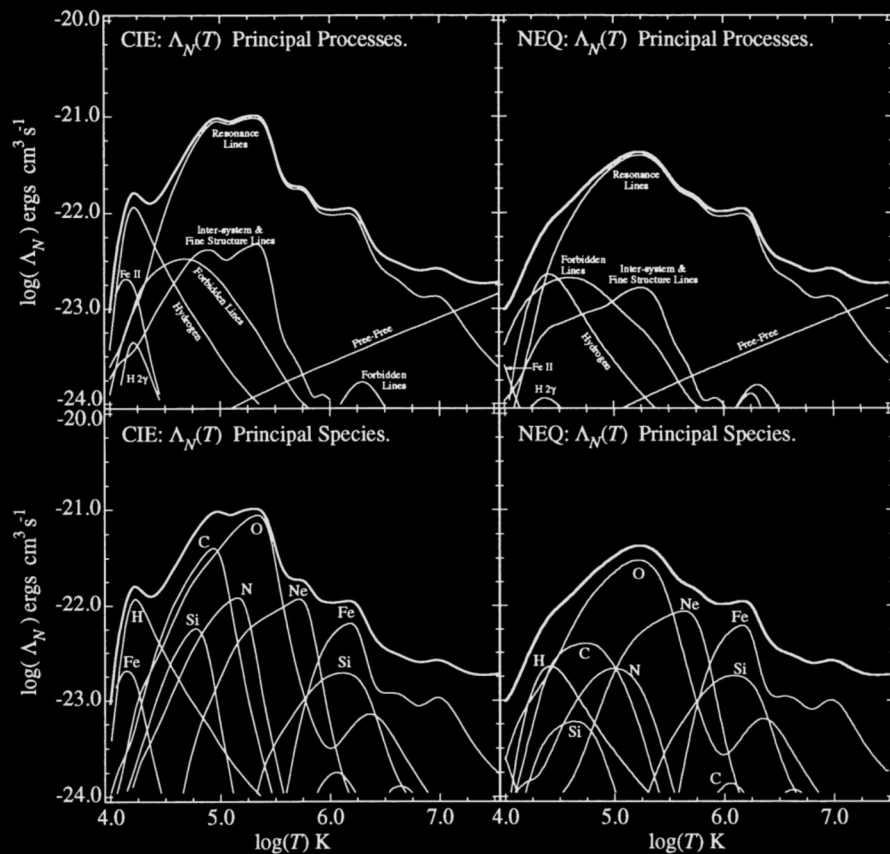
Appendix C1: Heating/Cooling Mechanisms

Heating	Cooling	
H ₂ formation	H ₂ ro-vibrational	H ₂
	H ₂ CIE	
	HD	
PdV heat	H ₂ dissociation	
	He ³	
	LiH	
	Compton atomic H	

Appendix C2: H₂ Cooling in Primordial Gas Clouds

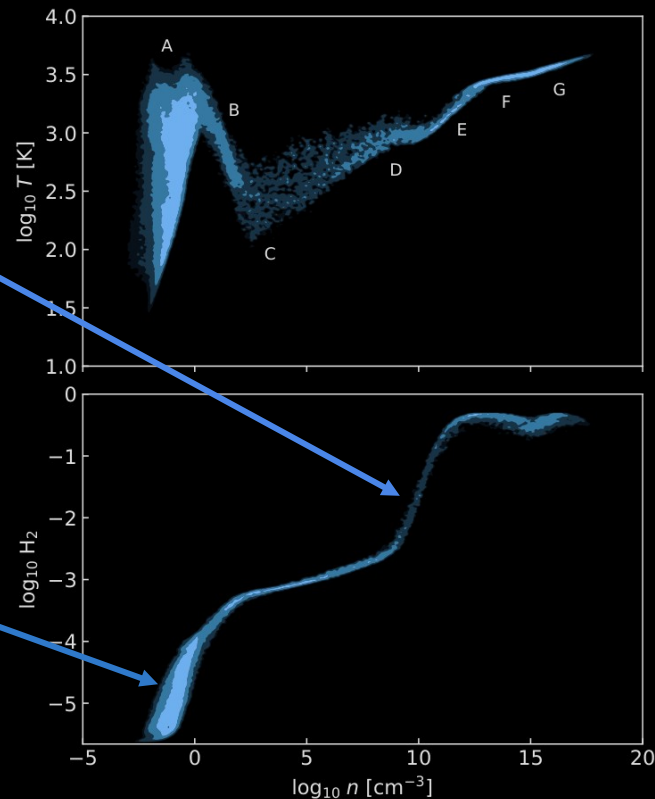
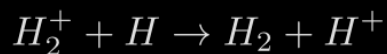
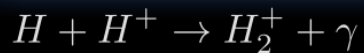
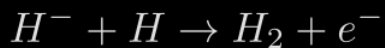
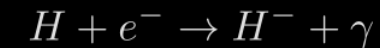
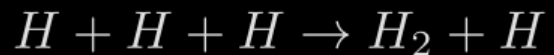


Appendix C3: Cooling Contribution of Different Species

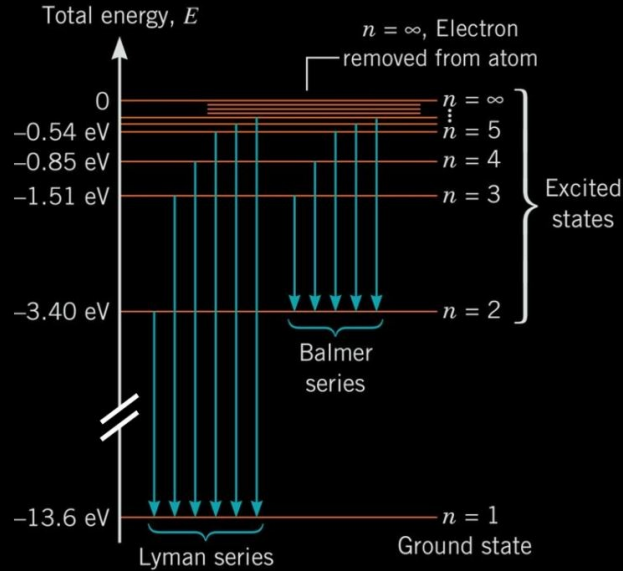


Credit:
Sutherland & Dopita, 1993

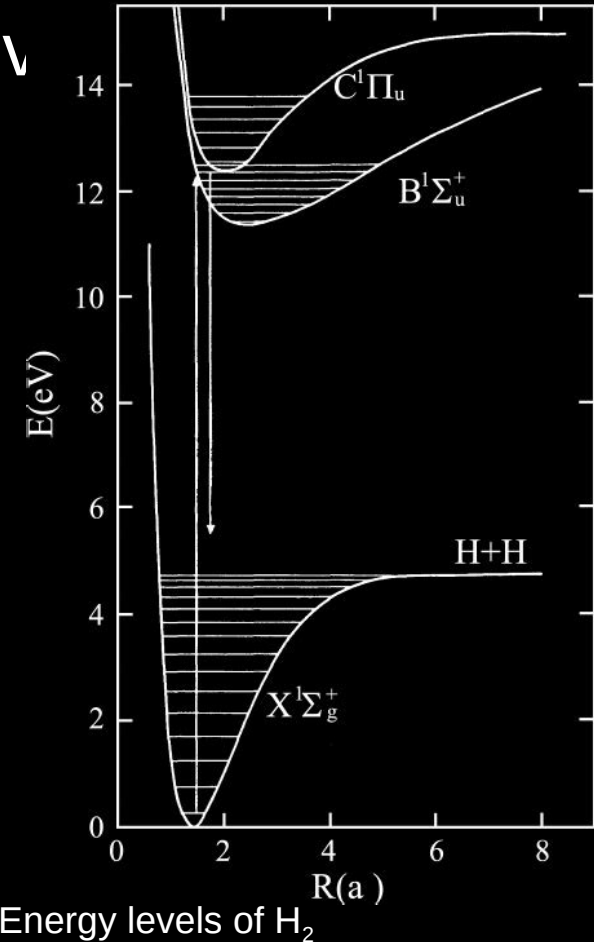
Appendix D: H₂ Formation Pathways



Appendix E: H and H₂ Energy lev



Energy level diagram of Hydrogen



Energy levels of H_2

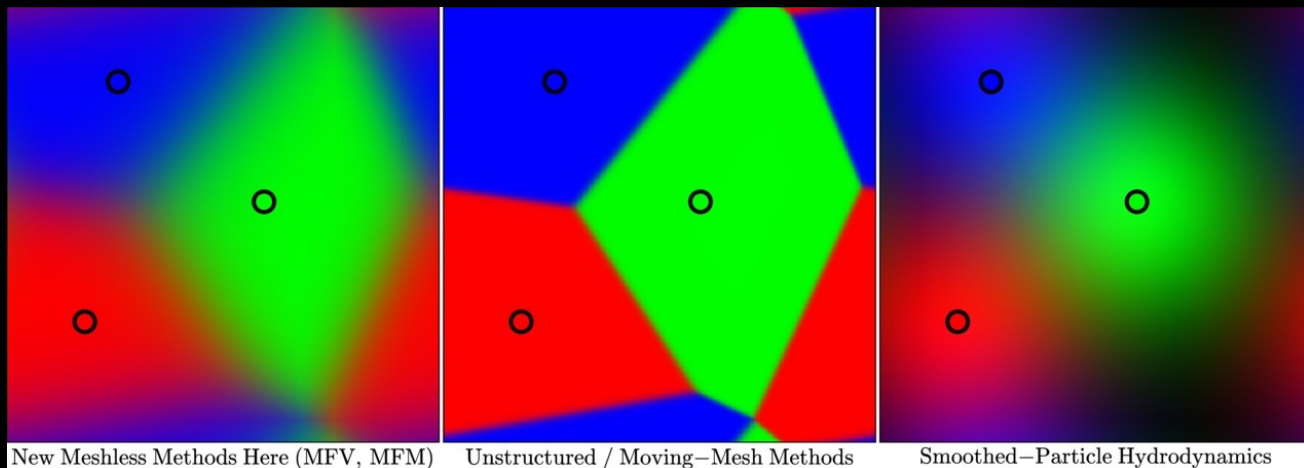
Appendix F: Cosmological Fluid Equation

$$\begin{aligned}
 & \frac{\partial \rho}{\partial t} + \vec{\nabla}_r \cdot \rho \vec{u} = 0 \\
 & \frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \vec{\nabla}_r) \vec{u} = -\frac{1}{\rho} \vec{\nabla}_r P - \vec{\nabla}_r \Phi \quad \Rightarrow \quad \frac{\partial^2 \delta}{\partial t^2} - c_s^2 \vec{\nabla}^2 \delta + 2H \frac{\partial \delta}{\partial t} = 4\pi G \rho_u \delta \\
 & \nabla_r^2 \Phi = 4\pi G \left(\rho + \frac{3P}{c^2} \right) \\
 & \lambda_J = \left(\frac{15 k_B T}{4\pi G \mu \rho} \right)^{\frac{1}{2}} \quad \delta(x, t) = \frac{\rho(x, t) - \langle \rho \rangle(t)}{\langle \rho \rangle(t)}
 \end{aligned}$$

Appendix G: Specifications of GIZMO

Unlike SPH, MFV/MFM uses kernel function only to partition the space

Unlike mesh methods, MFV/MFM is meshless in nature



GIZMO

AREPO

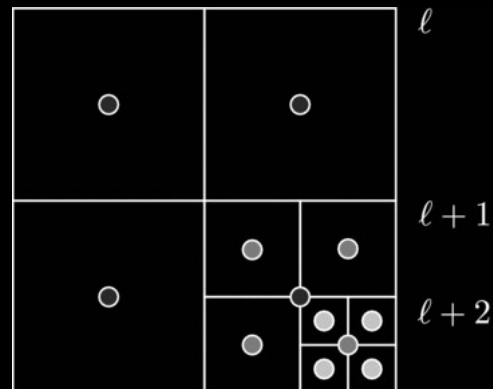
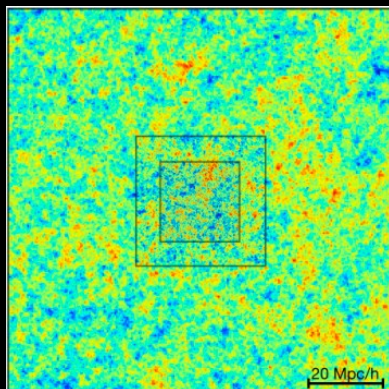
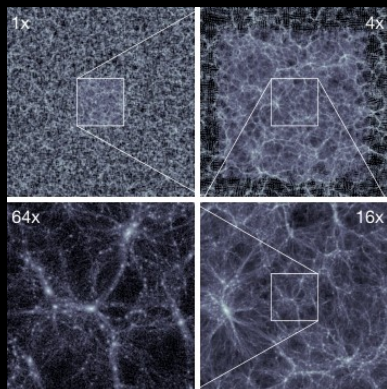
GADGET

Credit: GIZMO Users Guide

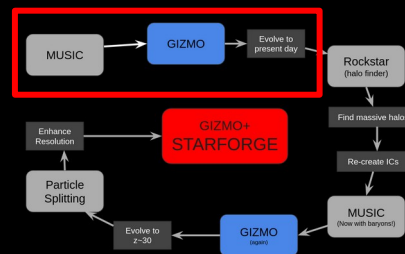
Appendix H: Multi-Scale Initial Conditions with MUSIC

Lowest refinement level - nested periodic DM grid

Highest refinement level - high-res DM and baryons

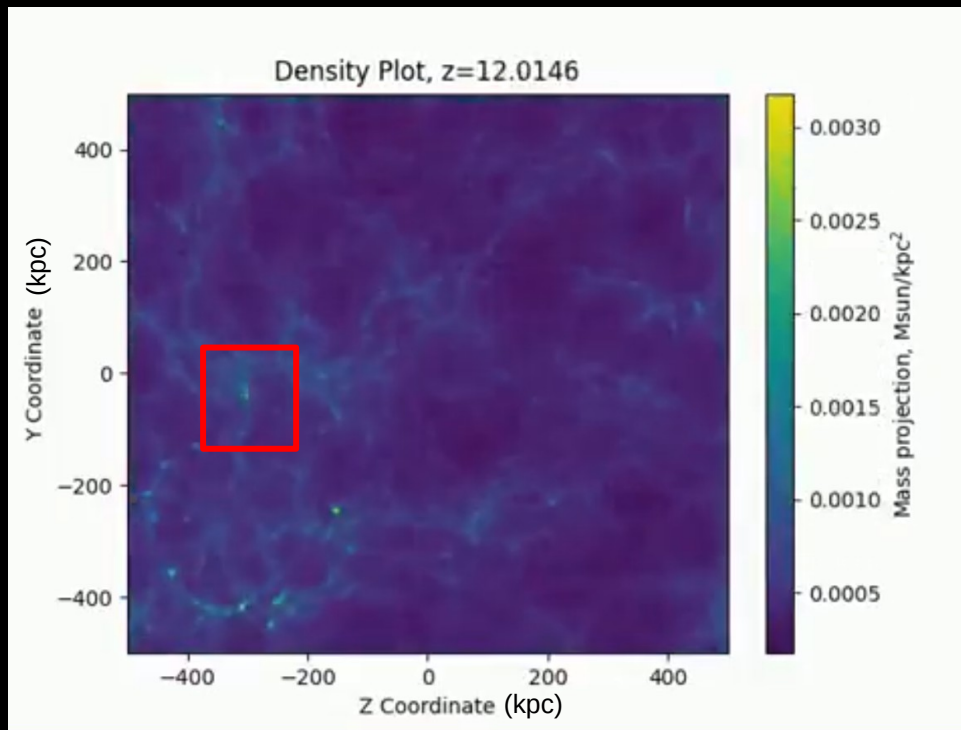
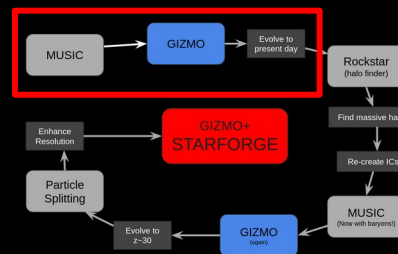


Simulation Process: Let's Simulate!

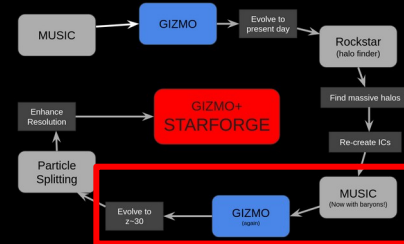
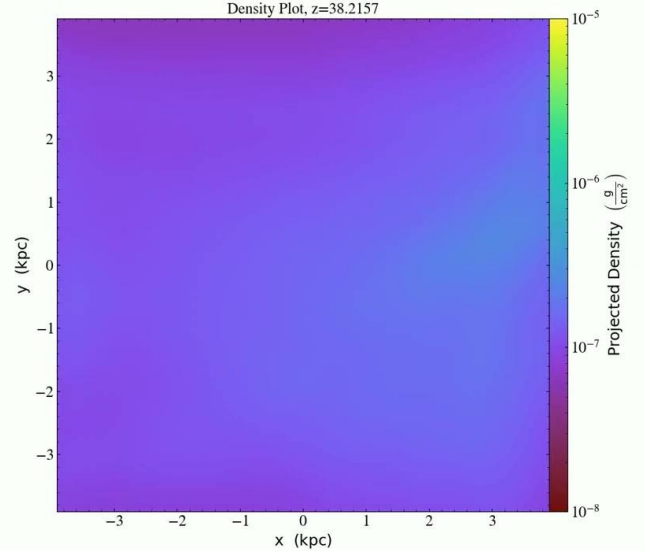
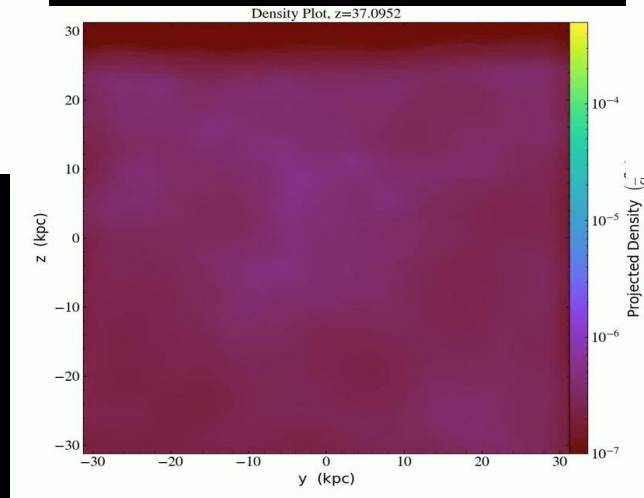
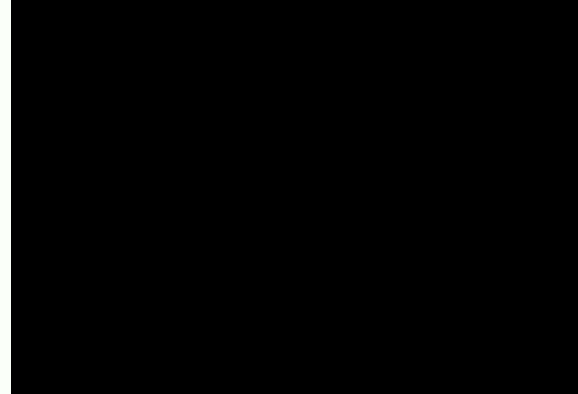
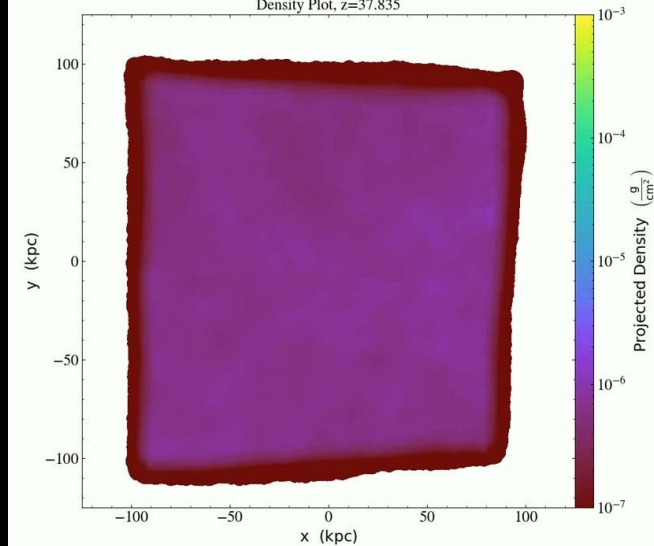


This is DM-only sim!
Coordinate units: kpc/h
Box size: 1 Mpc/h

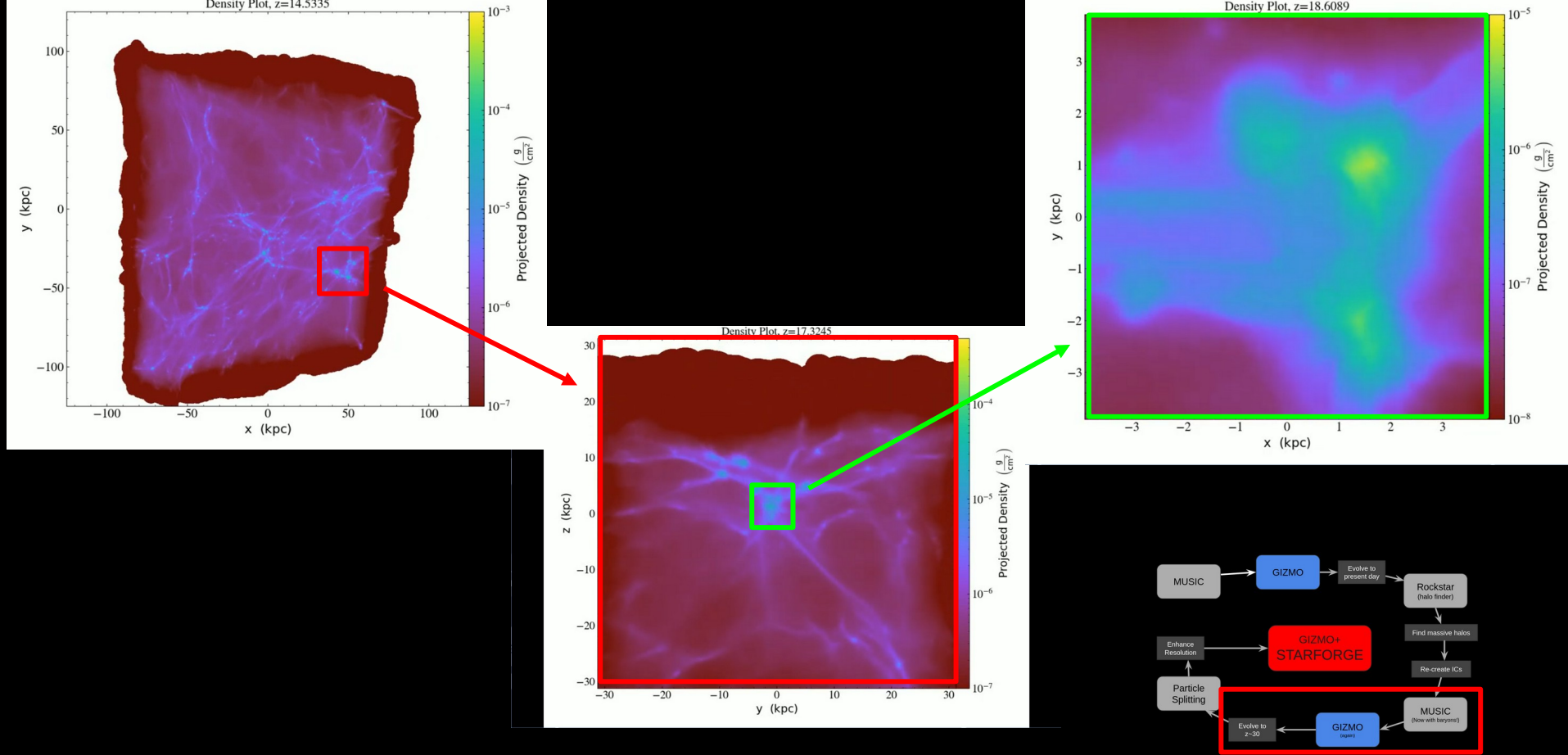
Simulation Process: Let's Simulate!



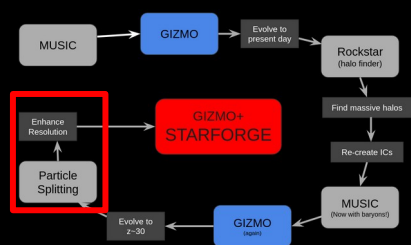
This is DM-only sim!
Coordinate units: kpc/h
Box size: 1 Mpc/h



This is DM+baryons sim!

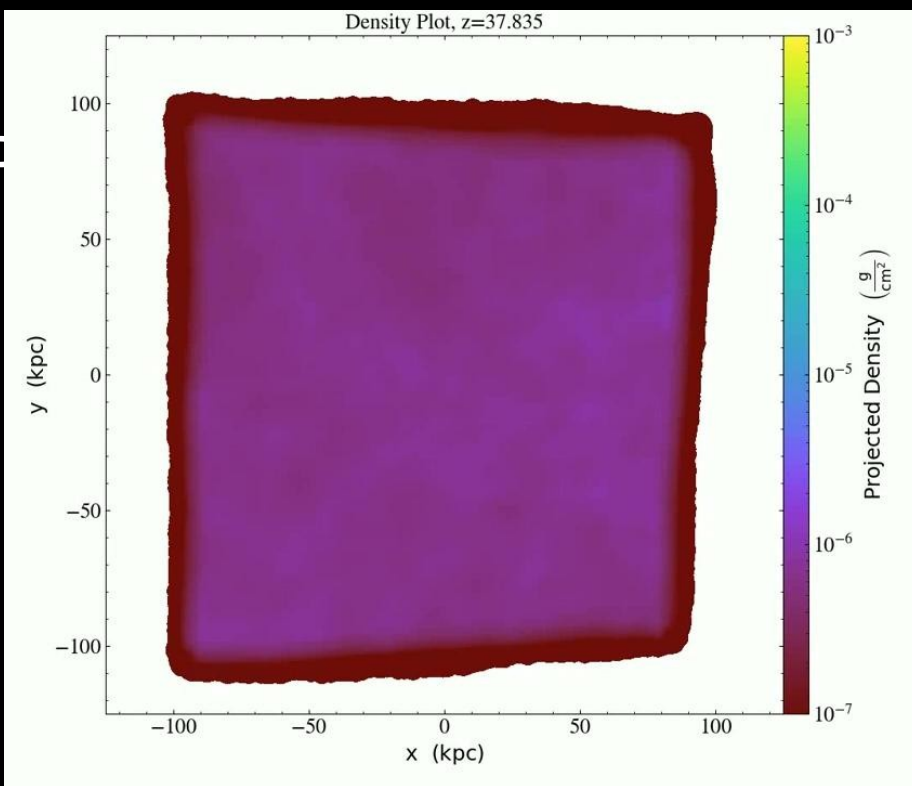


This is DM+baryons sim!



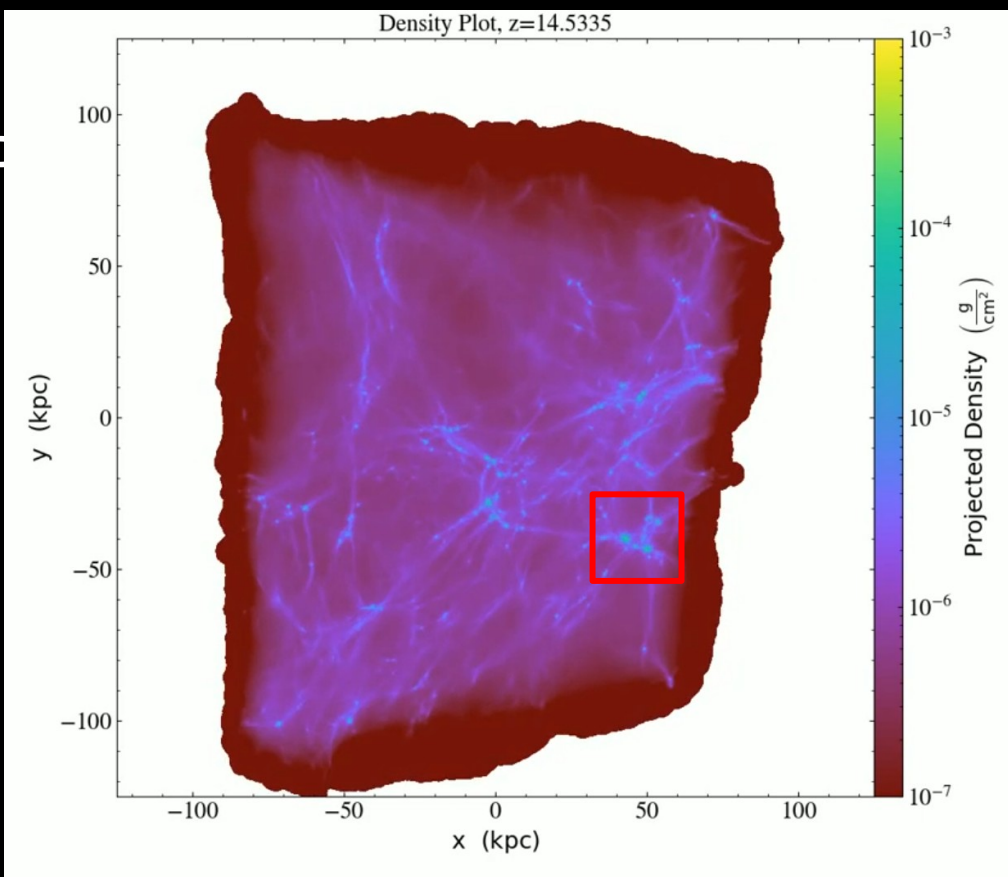
Failed attempts at particle splitting... (so far)

Simulation F



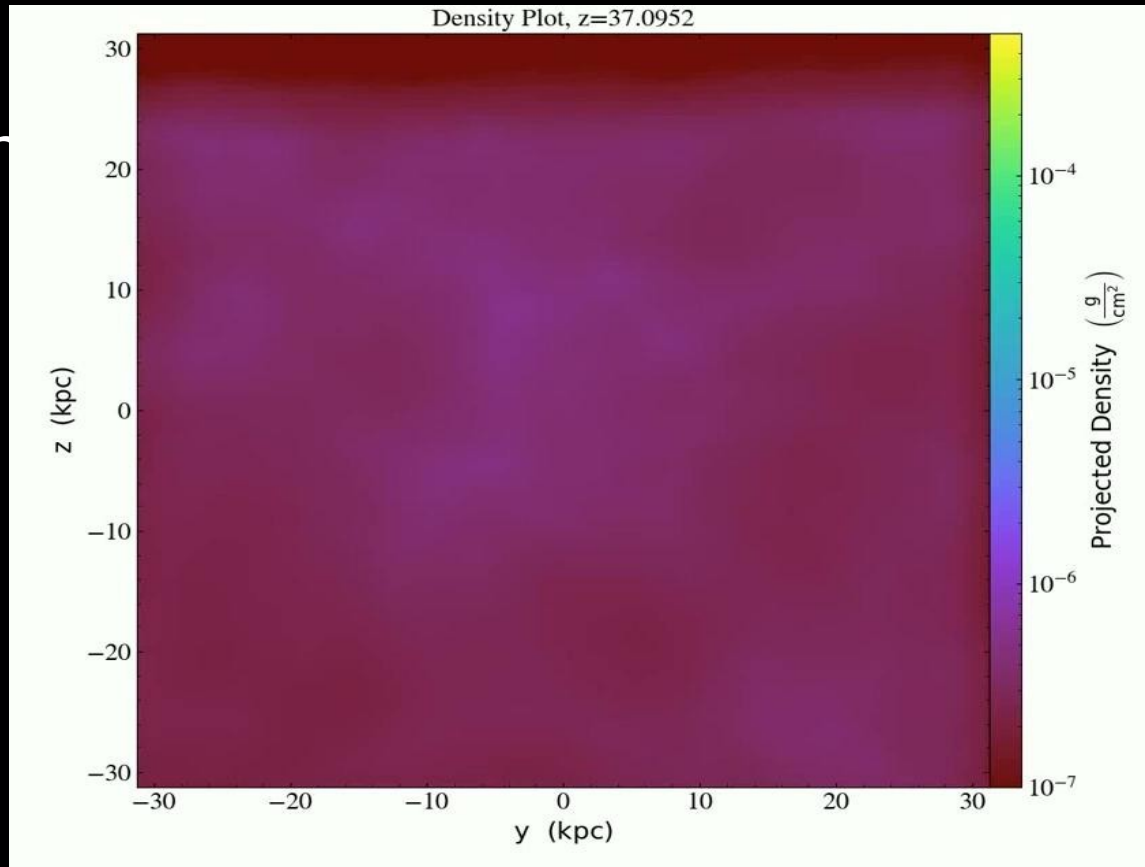
This is DM+baryons sim!

Simulation F



This is DM+baryons sim!

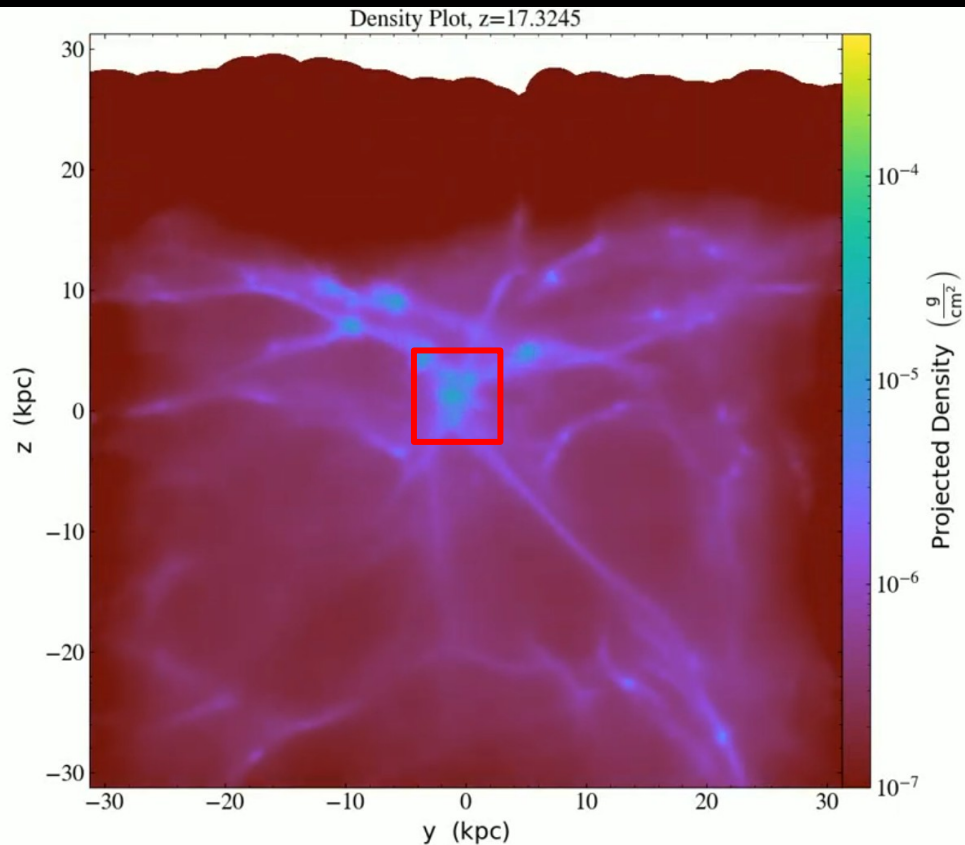
Simulation



This is DM+baryons sim!

x2

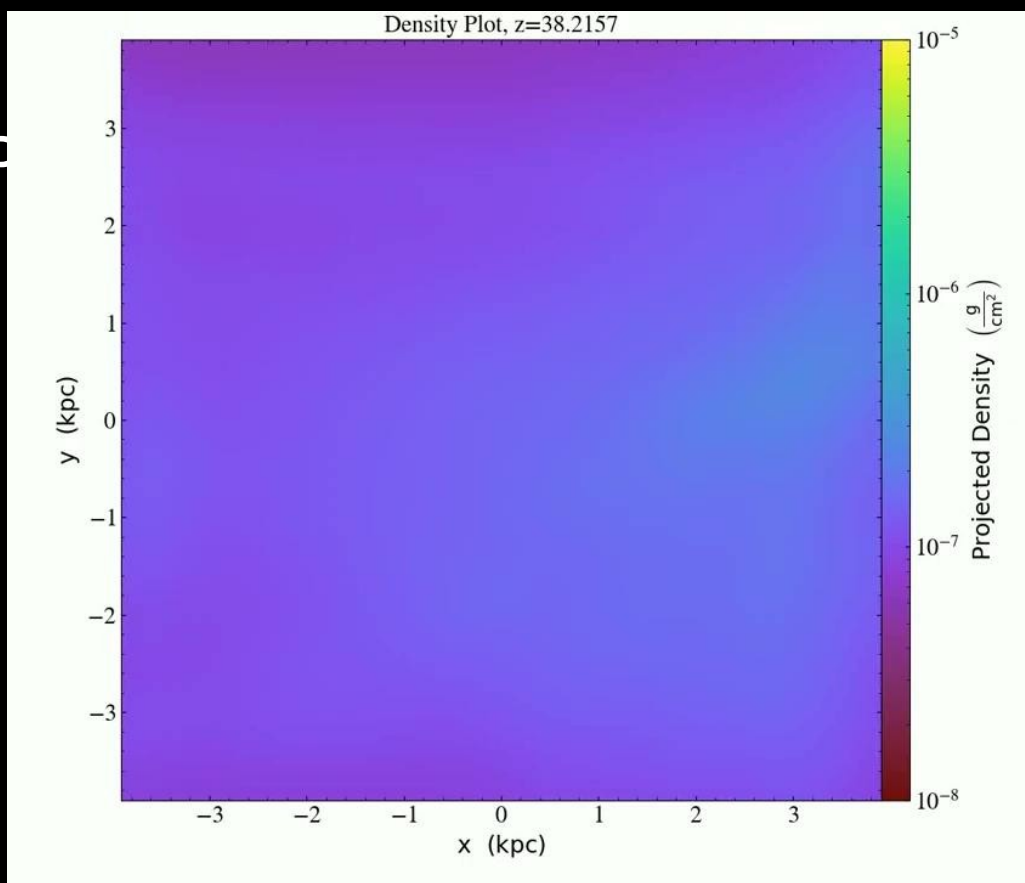
Simulation



This is DM+baryons sim!

x2

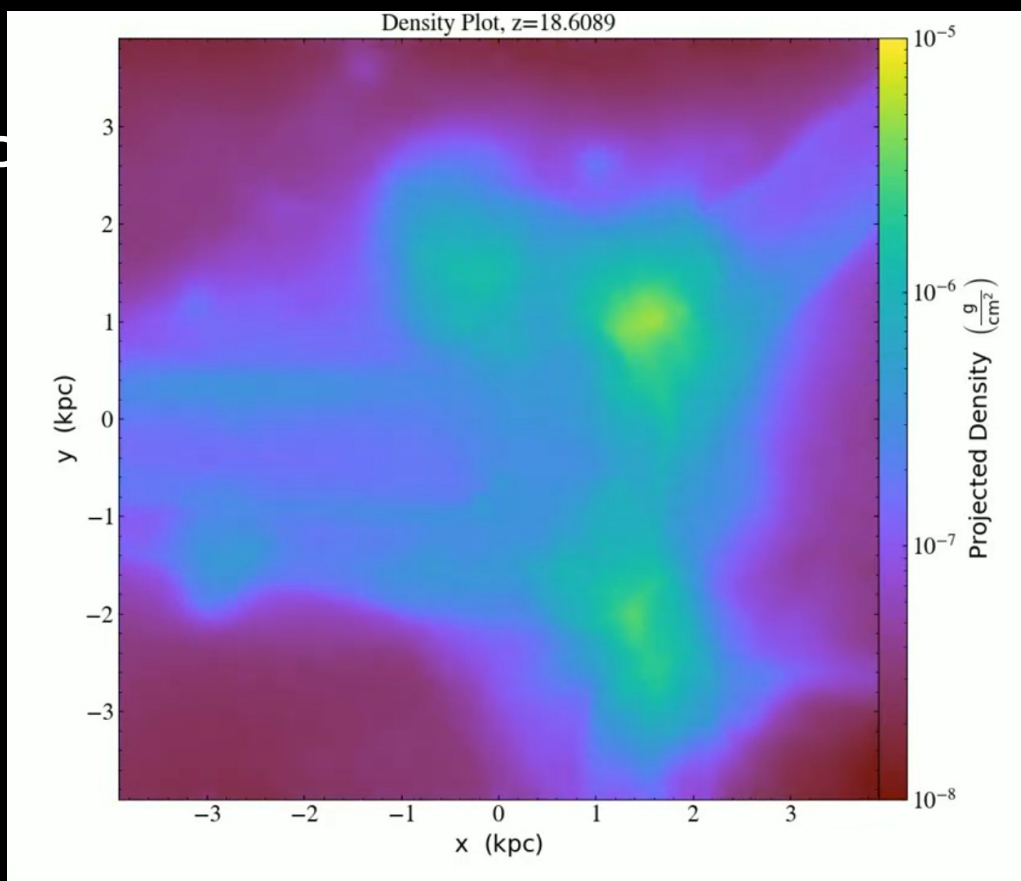
Simulation F



This is DM+baryons sim!

x3

Simulation F



This is DM+baryons sim!

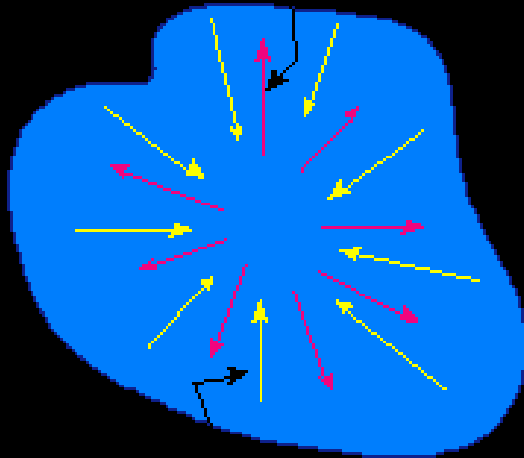
x3

It's not.*

*But what about composition of these clouds?

PopI Star Formation 101

Gas pressure attempting to
expand the cloud



Gravitational force attempting
to collapse the cloud

Pressure vs Gravity

Time for density perturbation
to travel across the star vs Free-fall time

For gravitational collapse to happen:

$$t_{ff} < t_{sound} \Leftrightarrow F_G > F_{pres}$$

$$\lambda_J = \left(\frac{15k_B T}{4\pi G \mu \rho} \right)^{\frac{1}{2}}$$

Gravity would dominate the collapse on scales
ABOVE the Jean's length!